Cisco 8600 Series Secure Routers Hardware Installation Guide

Cisco 8600 Secure Router Gen2
Updated October 12, 2026

© 2026 Cisco Systems, Inc. All rights reserved.

Full Cisco Trademarks with Hardware License

THE SPECIFICATIONS AND INFORMATION REGARDING THE PRODUCTS IN THIS MANUAL ARE SUBJECT TO CHANGE WITHOUT NOTICE. ALL STATEMENTS, INFORMATION, AND RECOMMENDATIONS IN THIS MANUAL ARE BELIEVED TO BE ACCURATE BUT ARE PRESENTED WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED. USERS MUST TAKE FULL RESPONSIBILITY FOR THEIR APPLICATION OF ANY PRODUCTS.

THE SOFTWARE LICENSE AND LIMITED WARRANTY FOR THE ACCOMPANYING PRODUCT ARE SET FORTH IN THE INFORMATION PACKET THAT SHIPPED WITH THE PRODUCT AND ARE INCORPORATED HEREIN BY THIS REFERENCE. IF YOU ARE UNABLE TO LOCATE THE SOFTWARE LICENSE OR LIMITED WARRANTY, CONTACT YOUR CISCO REPRESENTATIVE FOR A COPY.

The following information is for FCC compliance of Class A devices: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio-frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case users will be required to correct the interference at their own expense.

The following information is for FCC compliance of Class B devices: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If the equipment causes interference to radio or television reception, which can be determined by turning the equipment off and on, users are encouraged to try to correct the interference by using one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Modifications to this product not authorized by Cisco could void the FCC approval and negate your authority to operate the product.

The Cisco implementation of TCP header compression is an adaptation of a program developed by the University of California, Berkeley (UCB) as part of UCB's public domain version of the UNIX operating system. All rights reserved. Copyright © 1981, Regents of the University of California.

NOTWITHSTANDING ANY OTHER WARRANTY HEREIN, ALL DOCUMENT FILES AND SOFTWARE OF THESE SUPPLIERS ARE PROVIDED "AS IS" WITH ALL FAULTS. CISCO AND THE ABOVE-NAMED SUPPLIERS DISCLAIM ALL WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING, WITHOUT LIMITATION, THOSE OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OR ARISING FROM A COURSE OF DEALING, USAGE, OR TRADE PRACTICE.

IN NO EVENT SHALL CISCO OR ITS SUPPLIERS BE LIABLE FOR ANY INDIRECT, SPECIAL, CONSEQUENTIAL, OR INCIDENTAL DAMAGES, INCLUDING, WITHOUT LIMITATION, LOST PROFITS OR LOSS OR DAMAGE TO DATA ARISING OUT OF THE USE OR INABILITY TO USE THIS MANUAL, EVEN IF CISCO OR ITS SUPPLIERS HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

Any Internet Protocol (IP) addresses and phone numbers used in this document are not intended to be actual addresses and phone numbers. Any examples, command display output, network topology diagrams, and other figures included in the document are shown for illustrative purposes only. Any use of actual IP addresses or phone numbers in illustrative content is unintentional and coincidental.

All printed copies and duplicate soft copies of this document are considered uncontrolled. See the current online version for the latest version.

Cisco has more than 200 offices worldwide. Addresses and phone numbers are listed on the Cisco website at www.cisco.com/go/offices.

Cisco and the Cisco logo are trademarks or registered trademarks of Cisco and/or its affiliates in the U.S. and other countries. To view a list of Cisco trademarks, go to this URL: https://www.cisco.com/c/en/us/about/legal/trademarks.html. Third-party trademarks mentioned are the property of their respective owners. The use of the word partner does not imply a partnership relationship between Cisco and any other company. (1721R)

Topics included

Full Cisco Trademarks with Hardware License....ii

1 Introduction to Cisco 8600 Series Secure Routers....6

Cisco 8600 Series Secure Routers....7
Hardware features....7
AC power supply....8
DC power supply....8
Power supply fans....9
Chassis views....10
Bay configuration....12
LED indicators....13
Serial number and PID/VID label location....13

2 Preparing your site for installation....14

Prerequisites and preparation....15
Site planning checklist....15
Safety guidelines....16
Safety warnings....16
Safety recommendations....16
Cautions and regulatory compliance statements for NEBS....17
Electrical safety....19
Standard warning statements....20
Site planning....24
General precautions....24
Site cabling guidelines....24
Prevent electrostatic discharge damage....25
Rack-mounting guidelines....26
Precautions for rack-mounting....26
General rack-selection guidelines....26
Equipment rack guidelines....27
Chassis-lifting guidelines....28
Tools and equipment....28
Unpack and verify shipping contents....29
Checking the shipping container contents....29

3 Install 8600 Series Secure Routers....30

Install and connect....31
Safety instructions for installation....32

Guidelines for installation....32
Guidelines for rack installation....33
Install the router....35
Attach the front rack-mount brackets....35
Attach the rear rack-mount brackets....36
Mount the router in the rack....38
Attach the cable management bracket....38
Chassis ground connection....40
Connect the power cables....42
Connect the console port cables....43
Connect to the USB 3.0 console interface....43
Management ethernet port cable connection....44

4 Remove and Replace Field Replaceable Units....46
Install and remove AC power supplies....47
Install AC power supplies....47
Remove AC power supplies....47
Install and remove DC power supplies....48
Install DC input power supplies....48
Remove DC power supplies....51
Remove and replace USB 3.0 interface....51
Remove and replace the fans....51
Remove the fans for C8650-G2....52
Installation and removal of air filter....52
Install the air filter....53
Remove the air filter....54
Install and remove SFP and SFP+ modules....55
Laser safety guidelines....56
Remove small form pluggable modules....58
Remove and replace the battery....59
Remove the battery....59
Replace the battery....61
Access the battery in C8650-G2....62
Repack the router....65

1 Introduction to Cisco 8600 Series Secure Routers

Topics:

- Cisco 8600 Series Secure Routers
- Hardware features
- Chassis views
- Bay configuration
- LED indicators
- Serial number and PID/VID label location

Cisco 8600 Series Secure Routers

Introduces the Cisco 8600 Series Secure Routers, highlighting their primary functions, features, and intended deployment scenarios.

The Cisco 8600 Series Secure Routers deliver a robust and streamlined foundation for modern enterprise networking, supporting high-capacity VPN aggregation and high-density connectivity for large-scale, dataintensive operations and AI workloads.

This document covers installation specific details for the C8650-G2 model of the Cisco 8600 Series Secure Routers.

For more information on the features and specifications of Cisco 8600 Series Secure Routers, refer to the sections listed in this chapter.

Hardware features

Details the hardware features of Cisco 8600 Series Secure Routers, describing AC and DC power supply options and integrated power supply fans to support reliable hardware operation.

Table 1: Hardware Features for Cisco 8600 Series Secure Routers

Feature	C8650-G2
Ethernet Ports	20 x 1/10GE SFP+ ports 6 x 40/100GE QSFP28 ports
Rack Units	Three
System Memory (DRAM)	64GB DRAM (default)
Storage	480 GB SSD (default) eUSB bootflash 32GB
Management Interface	1x RJ-45 Out-Of-Band (OOB) management 1x RJ-45/1x Micro-USB Console
Console Port	1xRJ45, 1x USB Micro-USB
USB Ports	USB Type C
Rack Installation	Four post

Table 2: Supported Transceivers for Cisco 8600 Series Secure Routers

Feature	C8650-G2
Supported Transceivers	1G and 10G SFP+ modules 40G and 100G QSFP modules QSFP power classes up to Class 8

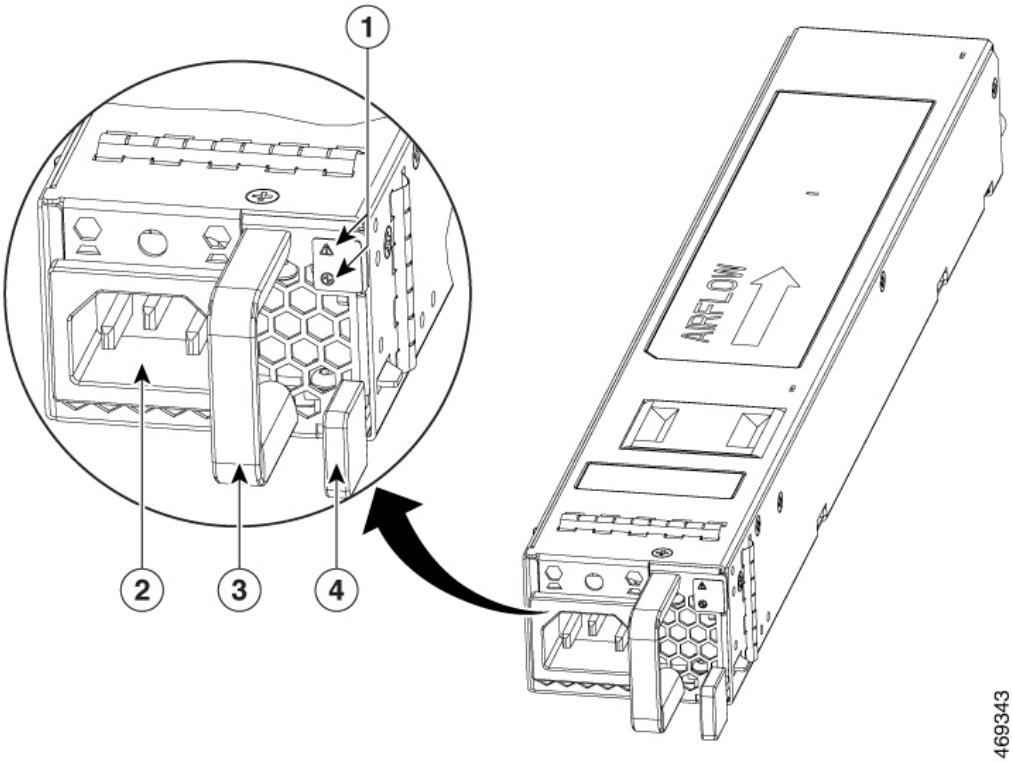
Table 3: Power Supply for Cisco 8600 Series Secure Routers

Power Supplies	C8650-G2
AC	PWR-CH1-1100WAC
DC	PWR-CH1-950WDC

AC power supply

This section describes the AC power supply for the Cisco 8600 Series Secure Routers.

Figure 1: AC Power Supply Used in the Cisco C8650-G2 Router



1	Fail and OK LEDs	3	Handle
2	AC power connector	4	Retaining latch

DC power supply

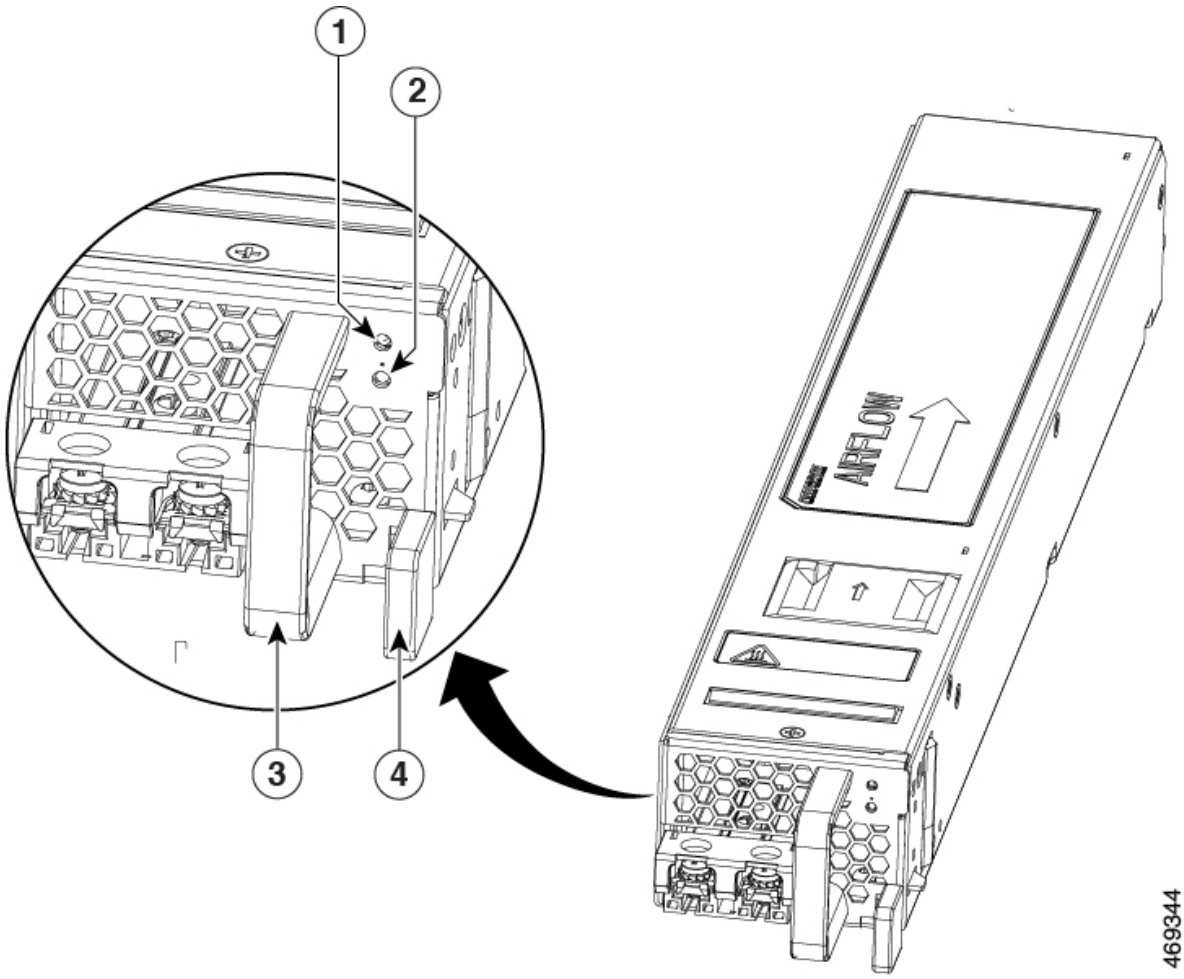
The DC input connector is a two-wire connector with connection polarity from left to right (when facing the unit) of positive (+) and negative (-).

The power supply has a handle to be used for insertion and extraction. The module must be supported with one hand because of its length.

Note

The airflow direction is front to back with ambient air drawn in from the venting located on the chassis front sides.

Figure 2: DC Power Supply for C8650-G2



1 and 2	Fail and OK LEDs	3	Handle
4	Retaining latch	–	

Power supply fans

The fans in the power supply module are used for cooling the power supply module itself while system-level cooling is provided by fans within the chassis. The power supplies do not depend on the system-level fans for cooling. Fan failure is determined by fan-rotation sensors.

Caution

The chassis has a front-to-rear airflow. All of the power supplies and fan modules in the same chassis must use the same airflow direction or an error will occur with possible overheating and shut down of the router. If you power up the router with more than one airflow direction, you must power down the router and replace the modules with the wrong airflow direction before powering up the router.

Note

The fans in the power supply modules will run as soon as the power supply is plugged in, even if the power switch is in the Standby position.

Chassis views

Outlines the chassis views of the Cisco 8600 Series Secure Routers, illustrating major hardware components and physical interfaces.

This section contains front and back panel views of the Cisco 8600 Series Secure Routers.

The different views of Cisco C8650-G2 chassis:

Figure 3: Cisco C8650-G2 Chassis—Front View

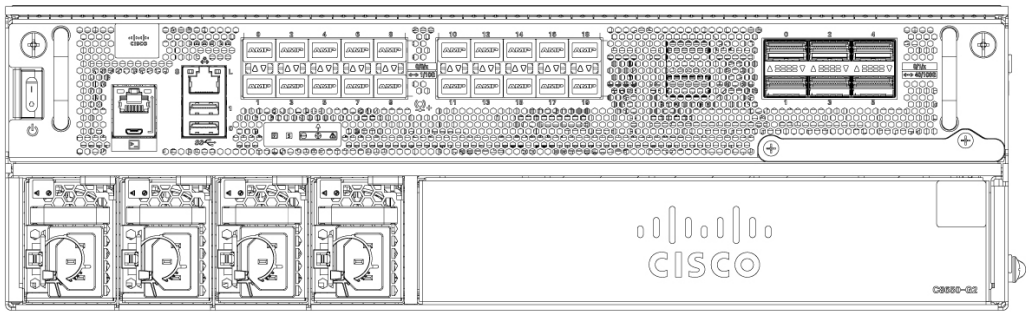
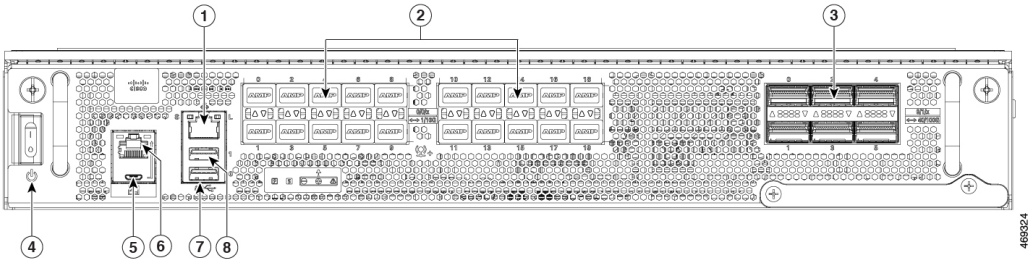


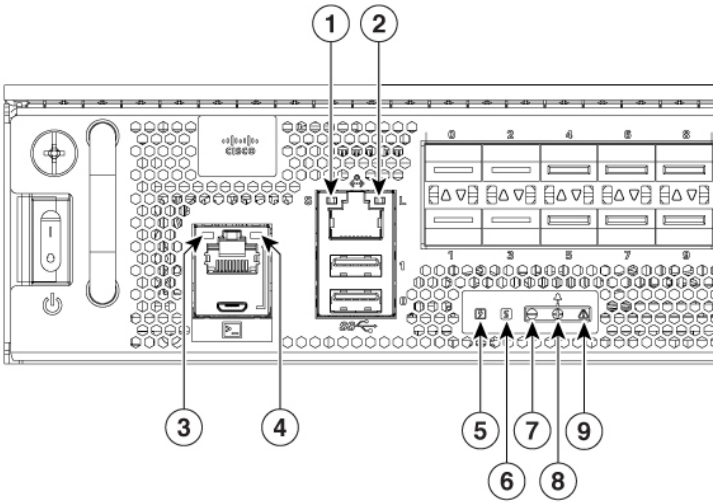
Figure 4: Cisco C8650-G2 Front View—Bays, Ports



1	Management Interface RJ-45	5	Micro-USB Console
2	Bay 0	6	Console RJ-45
3	Bay 1	7	USB 0

4	Power switch	8	USB 1

Figure 5: Cisco C8650-G2 Front View—LEDs



1	Management interface LED Flashing green : Link activity indicator Off : No Link Solid Green : Link with no activity	5	Power LED Off : No power to chassis. Yellow : Power On, one power supply has failed or is not plugged in. Green : All power is within specifications
2	Management interface LED - Link Link LED : Off - no link, Link LED : On - Link is up	6	Status LED Off : System not booted Red : System Failure Yellow : System is booted to ROMmon Green : System is booted to IOS
3	Micro-USB LED Left LED On indicates that USB console is active	7,8, and 9	Alarm LED : Minor, Major, Critical
4	Console RJ-45 LED Right LED On indicates that RJ-45 console is active		–

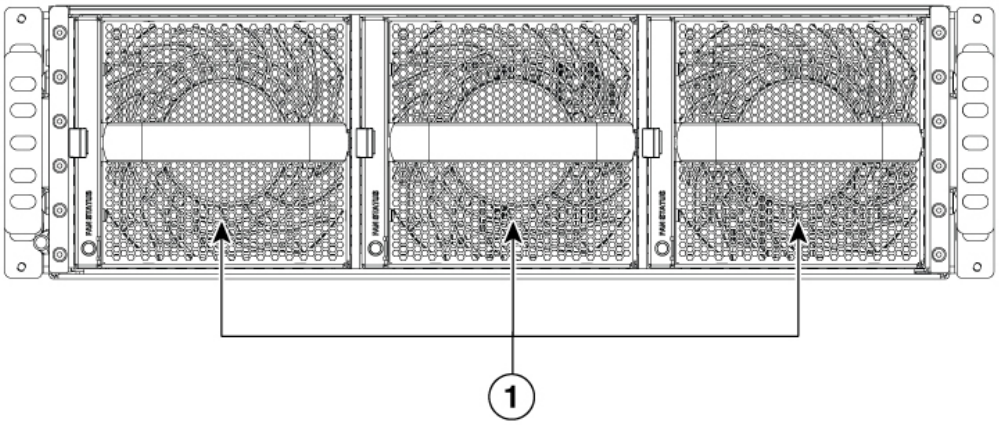
Figure 6: Cisco C8650-G2 Front View— AC Power Supply



Figure 7: Cisco C8650-G2 Front View— DC Power Supply



Figure 8: Cisco C8650-G2 Chassis—Rear View



1 Fans

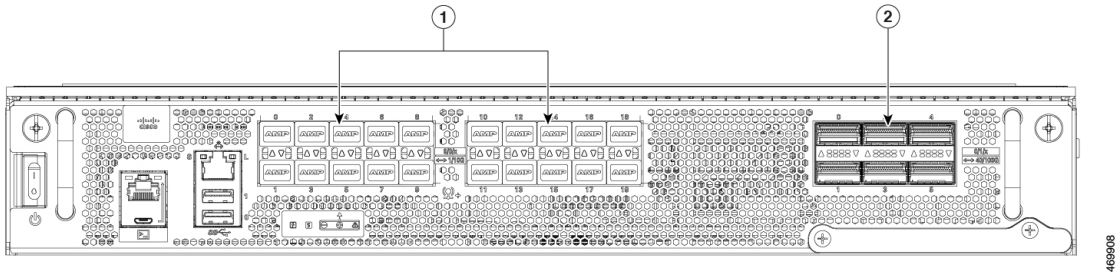
Bay configuration

Describes bay configuration options, guiding users on available expansion, slot layouts, and modularity features of the Cisco 8600 Series Secure Routers.

The Cisco 8600 Series Secure Routers are built for high-scale data center and colocation deployments.

The C8650-G2 router has two bays and provides 20x 10GE ports and 6x 100GE ports.

Figure 9: Bay Configuration - C8650-G2



1	Bay 0 : 20xSFP+ 20 ports that can be configured as : • 1G • 10G • a mix of 1G and 10G	2	Bay 1 : 6xQSFP+ 6 ports that can be configured as : • 40G • 100G • a mix of 40G and 100G

LED indicators

Explains the functions and interpretations of the LED indicators on Cisco 8600 Series Secure Routers for monitoring device status and activity.

This table describes the power supply LED.

Table 4: AC and DC Power Supply LED

Power Supply Condition	Green (OK) LED Status	Amber (FAIL) LED Status
No power to all power supplies	Off	Off
Power Supply Failure (includes over voltage, over current, over temperature and fan failure)	Off	On
Power Supply Warning events where the power supply continues to operate (high temperature, high power and slow fan)	Off	1Hz (blinking once per second)
AC Present/3.3VSB on (PSU Off)	1Hz (blinking once per second)	Off
Power Supply On and OK	On	Off

Serial number and PID/VID label location

Guides users in locating the serial number and PID/VID labels on Cisco 8600 Series Secure Routers for identification and inventory purposes.

On C8650-G2 router, the PID/VID label is located on top of the chassis.

2 Preparing your site for installation

Topics:

- Prerequisites and preparation
- Safety guidelines
- Site planning
- Rack-mounting guidelines
- Chassis-lifting guidelines
- Tools and equipment
- Unpack and verify shipping contents

Prerequisites and preparation

Introduces the essential prerequisites for installation readiness and provides a comprehensive site planning checklist to ensure proper preparation before deployment.

This chapter contains important safety information you should know before working with the Cisco 8600 Series Secure Routers and guides you through the process of preparing your site for router installation.

Before you perform the procedures in this guide, we recommend that you:

- Read the safety guidelines in the next section and review the electrical safety and ESD-prevention guidelines in this guide.
- Ensure that you have all of the necessary tools and equipment (see the "Tools and Equipment" section).
- Ensure that you have access to the *Cisco 8600 Series Secure Router Software Configuration Guide* (an online document that is available for viewing or download at Cisco.com) during the installation.
- Ensure that the power and cabling requirements are in place at your installation site.
- Ensure that the equipment required to install the router is available.
- Ensure that your installation site meets the environmental conditions to maintain normal operation.

Before installing the router, you must consider power and cabling requirements that must be in place at your installation site, special equipment for installing the router, and the environmental conditions your installation site must meet to maintain normal operation.

The shipping package for the router is engineered to reduce the chances of product damage associated with routine material handling experienced during shipment:

- Router should always be transported or stored in its shipping package in the upright position.
- Keep the router in the shipping container until you have determined the installation site.

> **Note**
>
> Inspect all items for shipping damage. If an item appears damaged, contact a Cisco customer service representative immediately.

Site planning checklist

Use this checklist to perform and account for all the site-planning tasks described in this chapter:

- The site air conditioning system can compensate for the heat dissipation of the router.
- Electrical service to the site complies with the requirements.
- The electrical circuit servicing the router complies with the requirements.
- Consideration has been given to console port wiring and limitations of the cabling involved, according to TIA/EIA-232F.
- The Ethernet cabling distances are within limitations.
- The equipment rack in which you plan to install the router chassis complies with requirements. Careful consideration has been given to safety, ease of maintenance, and proper airflow in selecting the location of the rack.

Safety guidelines

Details safety guidance for installation, including safety warnings, recommendations, NEBS compliance statements, electrical safety, and standard warning statements with general safety considerations.

Before you begin the installation or replacement procedure, review the safety guidelines in this section to avoid injuring yourself or damaging the equipment.

Note

This section contains guidelines, and do not include every potentially hazardous situation. When you install a router, always use common sense and caution.

Safety warnings

Safety warnings appear throughout this publication in procedures that, if performed incorrectly, might harm you. A warning symbol precedes each warning statement.

Before you install, configure, or perform maintenance on the router, review the documentation for the procedure you are about to perform, paying special attention to the safety warnings.

Note

Do not unpack the system until you are ready to install it. Keep the chassis in the shipping container to prevent accidental damage until you determine an installation site. Use the appropriate unpacking documentation included with the system.

Read the installation instructions in this document before you connect the system to its power source. Failure to read and follow these guidelines could lead to an unsuccessful installation and possibly damage the system and components.

Safety recommendations

These guidelines will help to ensure your own safety and protect your Cisco equipment. This list does not cover all potentially hazardous situations, so *be alert*.

- Cisco safety policy mandates that all its routers must conform to the requirements of IEC 60950-1 and IEC 62368-1, with appropriate national deviations, as a minimum. In addition, Cisco routers must also meet the requirements of any other normative documents, for example, standards, technical specifications, laws or regulations.
- Review the safety warnings listed in *Regulatory Compliance and Safety Information for the Cisco 8600 Series Secure Routers* (available online at Cisco.com) before installing, configuring, or maintaining the router.
- Never attempt to lift an object that might be too heavy for you to lift by yourself.
- Always turn all power supplies off and unplug all power cables before opening the chassis.
- Always unplug the power cable before installing or removing a chassis.
- Keep the chassis area clear and dust free during and after installation.
- Keep tools and chassis components away from walk areas.

- Do not wear loose clothing, jewelry (including rings and chains), or other items that could get caught in the chassis. Fasten your tie or scarf and sleeves.
- The router operates safely when it is used in accordance with its marked electrical ratings and product-usage instructions.

Cautions and regulatory compliance statements for NEBS

The table lists cautions, regulatory compliance statements, and requirements for the Network Equipment Building System (NEBS) certification.

Note

This equipment may be ESD sensitive. Always use an ESD ankle or wrist strap before handling equipment. Connect the equipment end of the ESD strap to an unfinished surface of the equipment chassis or to the ESD jack on the equipment if provided.

Note

The intrabuilding port(s) of the equipment or subassembly must use shielded intrabuilding cabling/wiring that is grounded at both ends. These port(s) are considered intrabuilding ports on this equipment:

- RJ-45 Copper Ethernet Ports

Note

To comply with the emission and immunity requirements of GR-1089, shielded cables are required for these ports:

- RJ-45 Copper Ethernet Ports

Note

The intrabuilding port(s) of the equipment or subassembly is suitable for connection to intrabuilding or unexposed wiring or cabling only. The intrabuilding port(s) of the equipment or subassembly MUST NOT be metallically connected to interfaces that connect to the OSP or its wiring for more than 6 meters (approximately 20 feet). These interfaces are designed for use as intrabuilding interfaces only (Type 2, 4, or 4a ports as described in GR-1089) and require isolation from the exposed OSP cabling. The addition of primary protectors is not sufficient protection in order to connect these interfaces metallically to an OSP wiring system. These ports are considered intrabuilding ports on the equipment:

- RJ-45 Copper Ethernet Ports

Warning

This product is intended to be protected by a surge protector that meets the applicable criteria of GR-974-CORE or GR-1361-CORE. Failure to use this appropriate surge protector could result in susceptibility to lightning surges or create a potential hazard due to power faults.

Warning

Connect this equipment to AC mains that are provided with a surge protective device (SPD) at the service equipment that complies with NFPA 70, the National Electrical Code (NEC).

Note

This equipment is suitable for installations using the CBN.

Note

When you use thread-forming screws to bond equipment to its mounting metalwork, remove any paint and nonconductive coatings and clean the joining surfaces. Apply an antioxidant compound before joining the surfaces between the equipment and mounting metalwork.

Note

Statement 7016– Battery Return Conductor

Treat the battery return conductor of this equipment as Isolated DC return (DC-I).

Note

The equipment is designed to boot up in less than 30 minutes provided the neighboring devices are fully operational.

Note

This equipment is suitable for installation in network telecommunications facilities.

Note

This equipment is suitable for installation in locations where the NEC applies.

Electrical safety

Follow these basic guidelines when you are working with any electrical equipment:

- Before beginning any procedures requiring access to the chassis interior, locate the emergency power-off switch for the room in which you are working.
- Disconnect all power and external cables before installing or removing a chassis.
- Do not work alone when potentially hazardous conditions exist.
- Never assume that power has been disconnected from a circuit; always check.
- Do not perform any action that creates a potential hazard to people or makes the equipment unsafe. Never install equipment that appears damaged.
- Carefully examine your work area for possible hazards such as moist floors, ungrounded power extension cables, and missing safety grounds.

Standard warning statements

Note

The English warnings in this document are preceded by a statement number. To see the translations of a warning in other languages, look up its statement number in the *Regulatory Compliance and Safety Information for the Cisco 8600 Series Secure Routers.*.

General safety warnings

The general safety warnings are as follows:

Warning

IMPORTANT SAFETY INSTRUCTIONS

Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Read the installation instructions before using, installing, or connecting the system to the power source. Use the statement number at the beginning of each warning statement to locate its translation in the translated safety warnings for this device.

SAVE THESE INSTRUCTIONS



Warning

Ultimate disposal of this product should be handled according to all national laws and regulations.

Warning

To reduce risk of electric shock or fire, installation of the equipment must comply with local and national electrical codes.

Warning

Statement 1005–Circuit Breaker

This product relies on the building's installation for short-circuit (overcurrent) protection. To reduce risk of electric shock or fire, ensure that the protective device is rated not greater than:

AC:

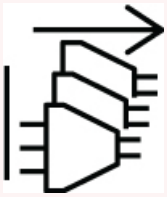
- 20 A U.S.

DC:

- 30 A U.S.

Warning

This unit might have more than one power supply connection. To reduce risk of electric shock, remove all connections to de-energize the unit.



Warning

This unit is intended for installation in restricted access areas. Only skilled, instructed, or qualified personnel can access a restricted access area.

Warning

Hazardous voltage or energy may be present on power terminals. To reduce the risk of electric shock, make sure the power terminal cover is in place when the power terminal is not being serviced. Be sure uninsulated conductors are not accessible when the cover is in place.

Warning

This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

Warning

Invisible laser radiation is present. Do not expose to users of telescopic optics. This applies to Class 1/1M laser products.



Warning

To prevent personal injury or damage to the chassis, never attempt to lift or tilt the chassis using the handles on modules, such as power supplies, fans, or cards. These types of handles are not designed to support the weight of the unit.

Warning

Blank faceplates and cover panels serve three important functions: they reduce the risk of electric shock and fire, they contain electromagnetic interference (EMI) that might disrupt other equipment, and they direct the flow of cooling air through the chassis. Do not operate the system unless all cards, faceplates, front covers, and rear covers are in place.

The laser safety warnings are as follows:

Warning

Invisible laser radiation may be emitted from disconnected fibers or connectors. Do not stare into beams or view directly with optical instruments.

Warning

Invisible laser radiation may be emitted from the end of the unterminated fiber cable or connector. Do not view directly with optical instruments. Viewing the laser output with certain optical instruments, for example, eye loupes, magnifiers, and microscopes, within a distance of 100 mm, may pose an eye hazard.

Fiber Type and Core Diameter (µm)	Wavelength (nm)	Maximum Power (mW)	Beam Divergence (rad)
SM 11	1200-1400	39-50	0.1-0.11
MM 62.5	1200-1400	150	0.18 NA
MM 50	1200-1400	135	0.17 NA
SM 11	1400-1600	112-145	0.11-0.13

> **Warning**
>
> Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.

Site planning

Explains site planning fundamentals, highlighting general precautions, site cabling guidelines–including USB serial console requirements–and procedures to prevent electrostatic discharge damage.

This section contains site-planning information, and will help you plan for the installation of the Cisco 8600 Series Secure Routers.

General precautions

Observe these general precautions when using and working with the Cisco 8600 Series Secure Routers

- Keep your system components away from radiators and heat sources and do not block cooling vents.
- Do not spill food or liquids on your system components and never operate the product in a wet environment.
- Do not push any objects into the openings of your system components. Doing so can cause fire or electric shock by shorting out interior components.
- Position system cables and power supply cable carefully. Route system cables and power supply cable and plug such that they cannot be stepped on or tripped over. Be sure that nothing else rests on your system component cables or power cable.
- Do not modify power cables or plugs. Consult a licensed electrician or your power company for site modifications. Always follow your local and national wiring rules.
- If you turn off your system, wait at least 30 seconds before turning it on again to avoid system component damage.

Site cabling guidelines

This section contains guidelines for wiring and cabling at your site. When preparing your site for network connections to the Cisco 8600 Series Secure Routers , consider the type of cable required for each component, and the cable limitations. Consider the distance limitations for signaling, EMI, and connector compatibility. Possible cable types are fiber, thick or thin coaxial, foil twisted-pair cabling, or unshielded twisted-pair cabling.

Also consider any additional interface equipment you need, such as transceivers, hubs, switches, modems, channel service units (CSUs), or data service units (DSUs).

Before you install the Cisco 8600 Series Secure Routers , have all the additional external equipment and cables at hand. For ordering information, contact a Cisco customer service representative.

The extent of your network and the distances between network interface connections depend in part on these factors:

- Signal type
- Signal speed
- Transmission medium

The distance and rate limits referenced in the next sections are the IEEE-recommended maximum speeds and distances for signaling purposes. Use this information as guidelines when planning your network connections prior to installing the Cisco 8600 Series Secure Routers.

If wires exceed recommended distances, or if wires pass between buildings, give special consideration to the effect of a lightning strike in your vicinity. The electromagnetic pulse caused by lightning or other high-energy phenomena can easily couple enough energy into unshielded conductors to destroy electronic devices. If you have had problems of this sort in the past, you may want to consult experts in electrical surge suppression and shielding.

USB serial console

The USB serial console port connects directly to the USB connector of a PC using a USB Type A to 5-pin mini USB Type-B cable. The USB Console supports full speed (12Mbps) operation. The console port does not support hardware flow control.

Note

- Always use shielded USB cables with a properly terminated shield. The USB serial console interface cable must not exceed 3 meters in length.
- Only one console port can be active at a time. When a cable is plugged into the USB console port, the RJ-45 port becomes inactive. Conversely, when the USB cable is removed from the USB port, the RJ-45 port becomes active.
- 4-pin micro USB Type-B connectors are easily confused with 5-pin micro USB Type-B connectors. Only 5-pin micro USB Type-B is supported.

Prevent electrostatic discharge damage

Electrostatic discharge (ESD) damage occurs when electronic cards or components are improperly handled resulting in complete or intermittent failures. Static electricity can harm delicate components inside your system. To prevent static damage, discharge static electricity from your body before you touch any of your system components, such as a microprocessor. As you continue to work on your system, periodically touch an unpainted metal surface on the computer chassis.

These are the guidelines for preventing ESD damage:

- Always use an ESD-preventive wrist or ankle strap and ensure that it makes good skin contact. Before removing a card from the chassis, connect the equipment end of the strap to the ESD plug at the bottom of the chassis below the power entry modules.
- Handle line cards by faceplates and carrier edges only; avoid touching the card components or connector pins.
- When removing a module, place the removed module component-side-up on an antistatic surface or in a static-shielding bag. If the module is to be returned to the factory, immediately place it in a static-shielding bag.
- Avoid contact between the modules and clothing. The wrist strap protects the card from ESD voltages only on the body; ESD voltages on clothing can still cause damage.
- When transporting a sensitive component, place it in an antistatic container or packaging.
- Handle all sensitive components in a static-safe area. If possible, use antistatic floor pads and workbench pads.

> ⚠ **Caution**
>
> For safety, periodically check the resistance value of the antistatic strap. The measurement should be less than 35 megohms.

> ⚠ **Caution**
>
> Always tighten the captive installation screws on all the system components when you are installing them. These screws prevent accidental removal of the module, provide proper grounding for the system, and help ensure that the bus connectors are properly seated in the backplane.

Rack-mounting guidelines

Outlines best practices for rack-mounting, covering necessary precautions, rack selection, and equipment placement to ensure safety, ease of maintenance, and proper airflow.

This section describes guidelines on rack-mounting.

Precautions for rack-mounting

The rack-mounting guidelines are provided to ensure your safety:

- Do not move large racks by yourself. Due to the height and weight of a rack, a minimum of two people are required to accomplish this task.
- Ensure that the rack is level and stable before extending a component from the rack.
- Ensure that proper airflow is provided to the components in the rack.
- Do not step or stand on any component or system when servicing other systems or components in a rack.
- When mounting the Cisco 8600 Series Secure Router in a partially filled rack, load the rack from the bottom to the top with the heaviest component at the bottom of the rack.
- If the rack is provided with stabilizing devices, install the stabilizers before mounting or servicing the unit in the rack.

General rack-selection guidelines

The Cisco 8600 Series Secure Router can be mounted in four-post, 19-in. equipment racks that comply with the Electronics Industries Association (EIA) standard for equipment racks (EIA-310-D 19-in.). The rack must have at least four posts with mounting flanges to mount the chassis.

Caution

When mounting a chassis in any type of rack equipment, ensure that the inlet air to the chassis does not exceed 40° C.

The distance between the center lines of the mounting holes on the two mounting posts must be 18.31 in. ± 0.06 in. (46.50 cm ± 0.15 cm). The rack-mounting hardware included with the chassis is suitable for most 19-in. (48.3-cm) equipment racks.

Consider installing the Cisco 8600 Series Secure Router in a rack with these features:

- NEBS-compliant, 19-in. (48.3-cm) wide rack.
- EIA or ETSI hole patterns in the mounting rails. Required mounting hardware is shipped with the Cisco 8600 Series Secure Router. If the rack that you plan to install the system in has metric-threaded rails, you must provide your own metric-mounting hardware.
- Perforated top and open bottom for ventilation to prevent overheating.
- Leveling feet for stability.

Note

The Cisco 8600 Series Secure Router should *not* be installed in an enclosed rack because the chassis requires an unobstructed flow of cooling air to maintain acceptable operating temperatures for its internal components. Installing the router in any type of enclosed rack–*even with the front and back doors removed*–could disrupt the air flow, trap heat next to the chassis, and cause an overtemperature condition inside the router. If you use an enclosed rack, make certain that there are air vents on all sides of the rack and there is proper ventilation.

Equipment rack guidelines

The placement of racks can affect personnel safety, system maintenance, and the system's ability to operate within the environmental characteristics described in Cisco 8600 Series Secure Routers. Choose a proper location for the Cisco 8600 Series Secure Routers by checking the guidelines listed below.

Locate for safety

If the Cisco 8600 Series Secure Routers is the heaviest or the only piece of equipment in the rack, consider installing it at or near the bottom to ensure that the rack's center of gravity is as low as possible.

Locate for easy maintenance

Keep at least 3 feet of clear space in front of and behind the rack. This space ensures that you can remove the Cisco 8600 Series Secure Routers components and perform routine maintenance and upgrades easily.

Avoid installing the Cisco 8600 Series Secure Routers in a congested rack and consider how the routing of cables from other pieces of equipment in the same rack might affect access to the routers cards.

The front and top of the chassis must remain unobstructed to ensure adequate airflow and prevent overheating inside the chassis.

Allow these clearances for normal system maintenance:

- At the top of the chassis–At least 3 in. (7.6 cm)
- In front of the chassis–3 to 4 ft (91.44 cm to 121.92 cm)

To avoid problems during installation and ongoing operation, follow these general precautions when you plan the equipment locations and connections:

- Use the **show environment all** and the **show facility-alarm status** commands regularly to check the internal system status. The environmental monitor continually checks the interior chassis environment; it provides warnings for high temperature and creates reports on any occurrences. If warning messages are displayed, take immediate action to identify the cause and correct the problem.
- Keep the Cisco 8600 Series Secure Routers off the floor and out of the areas that collect dust.
- Follow ESD-prevention procedures to avoid damage to equipment. Damage from static discharge can cause immediate or intermittent equipment failure.

Locate for proper airflow

Ensure that the location of the Cisco 8600 Series Secure Routers have enough airflow to keep the system operating within the environmental characteristics, and the air temperature is sufficient to compensate for the heat dissipated by the system.

Avoid locating the Cisco 8600 Series Secure Routers in a location in which the chassis air intake vents could draw in the exhaust air from adjacent equipment. Consider how the air flows through the router. The airflow direction is front to back with ambient air drawn in from the venting located on the chassis' front sides.

Chassis-lifting guidelines

Describes guidelines for safe chassis lifting during installation to prevent injury and equipment damage.

The chassis is not intended to be moved frequently. Before you install the system, ensure that your site is properly prepared so that you can avoid having to move the chassis later to accommodate power sources and network connections.

Each time you lift the chassis or any heavy object, follow these guidelines:

- Ensure that your footing is solid, and balance the weight of the chassis between your feet.
- Lift the chassis slowly; never move suddenly or twist your body as you lift.
- Keep your back straight and lift with your legs, not your back. If you must bend down to lift the chassis, bend at the knees, not at the waist, to reduce the strain on your back muscles.
- Do not remove installed components from the chassis.
- Always disconnect all external cables before lifting or moving the chassis.

Tools and equipment

Lists the tools and equipment required for the installation process to ensure readiness and efficiency.

These tools and equipment are recommended as the minimum necessary equipment to install the Cisco 8600 Series Secure Routers. You may need additional tools and equipment to install associated equipment and cables. You may also require test equipment to check electronic and optical signal levels, power levels, and communications links.

- Phillips hand screwdriver
- 3.5-mm flat-blade screwdriver
- Tape measure (optional)
- Level (optional)
- Power drill

- 8-gauge wire
- Rack-mount brackets
- Cable-management brackets

Unpack and verify shipping contents

Guides the unpacking process and verification of shipping container contents to ensure all necessary components are present before installation.

When you receive your chassis, perform these steps and use the shipping contents checklist in the next section.

Procedure

1. Inspect the box for any shipping damage. (If there is damage, contact your Cisco service representative).
2. Unpack the Cisco 8600 Series Secure Routers.
3. Perform a visual inspection of the chassis.
4. After you have unpacked the system, verify that you have received all of the required components, including all the accessory items. Using the packing list as a guide, verify that you have received all the equipment listed in your order, and ensure that the configuration matches the packing list.

Checking the shipping container contents

Use the components list shown in this table to check the contents of the Cisco 8600 Series Secure Routers shipping container. Do not discard the shipping container. You need the container if you move or have to ship the Cisco 8600 Series Secure Routers in the future.

Table 5: Cisco 8600 Series Secure Routers Shipping Container Contents

Component	Description
Chassis	Cisco 8600 Series Secure Routers are configured with four AC or four DC power supplies.
Accessories Kit (C8500-ACCKIT3R-19)	It contains: • Two front rack-mount brackets (6 screws for each bracket) • Two rear rack-mount brackets (4 screws for each bracket) • Two extension brackets • Two cable-management brackets (2 mounting screws for front bracket on each side)
Documentation	Pointer Doc
Optional Equipment	Power cord if an AC power supply was shipped. There are no power cords for the DC power supply units.

3 Install 8600 Series Secure Routers

Topics:

- Install and connect
- Install the router
- Connect the power cables

Install and connect

Introduces safety considerations and best practices for installing 8600 Series Secure Routers, detailing standalone and rack installation guidelines, including verification of rack dimensions to ensure proper equipment fit and secure deployment.

This chapter provides procedures for installing the Cisco 8600 Series Secure router on an equipment shelf, tabletop, or in an equipment rack. It describes how to install and connect Cisco 8600 Series Secure Routers to LAN and WAN networks.

Safety instructions for installation

Warning

IMPORTANT SAFETY INSTRUCTIONS

Before you work on any equipment, be aware of the hazards involved with electrical circuitry and be familiar with standard practices for preventing accidents. Read the installation instructions before using, installing, or connecting the system to the power source. Use the statement number at the beginning of each warning statement to locate its translation in the translated safety warnings for this device.

SAVE THESE INSTRUCTIONS



Note

IMPORTANT SAFETY INSTRUCTIONS

Invisible laser radiation may be emitted from the fiber cable or connector. Do not view with optical instruments. Viewing the laser output with certain optical instruments (for example, eye loupes, magnifiers, and microscopes) within a distance of 100 mm may pose an eye hazard.

Note

IMPORTANT SAFETY INSTRUCTIONS

Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.

Warning

Read the installation instructions before using, installing, or connecting the system to the power source.

Note

Proceed with the installation if you have already unpacked your chassis and read all the site requirements for your new equipment.

Guidelines for installation

The chassis should already be in the area where you want to install it. If you have not determined where to install your chassis,

- Keep at least 3 feet of clear space in front of and behind the rack. This space ensures that you can remove the Cisco 8600 Series Secure Router components and perform routine maintenance and upgrades easily.

  Avoid installing the Cisco 8600 Series Secure Router in a congested rack and consider how the routing of cables from other pieces of equipment in the same rack might affect access to the routers cards.

  The front and top of the chassis must remain unobstructed to ensure adequate airflow and prevent overheating inside the chassis.

  Allow these clearances for normal system maintenance:

  - At the top of the chassis–At least 3 in. (7.6 cm)
  - In front of the chassis–3 to 4 ft (91.44 cm to 121.92 cm)
  - Rear of the chassis–atleast 12 in.

- The chassis should be installed off the floor. Dust that accumulates on the floor is drawn into the interior of the router by the cooling fans. Excessive dust inside the router can cause overtemperature conditions and component failures.
- Maintain a minimum clearance of 5.25 inches (13.335 or 13.34 cm) on the front and back sides of the chassis for the cooling air inlet and exhaust ports, respectively. Avoid placing the chassis in an overly congested rack or directly next to another equipment rack; the heated exhaust air from other equipment can enter the inlet air vents and cause an overtemperature condition inside the router.
- The chassis needs adequate ventilation. Do not install it in an enclosed cabinet where ventilation is inadequate.
- Keep the cable-management bracket ready if you plan to install it on the front of the chassis.
- Ensure that an adequate chassis ground (earth) connection exists for your router chassis (see the "Attaching a Chassis Ground Connection" section).

Guidelines for rack installation

The Cisco 8600 Series Secure Routers can be installed in Four-post, 19-inch equipment rack. Inner clearance (the width between the inner sides of the two posts or rails) must be at least 19 inches (48.26 cm). Airflow through the chassis is from front to back.

The Cisco 8600 Series Secure Routers must be installed with both front and rear rack-mount brackets.

When planning your rack installation, consider these guidelines:

- The Cisco 8600 Series Secure Routers requires a minimum of 1.75 inches or 4.45 cm rack units of vertical rack space. The Cisco 8600 Series Secure Routers requires a minimum of 3.5 inches or 8.9 cm rack units of vertical rack space. Measure the proposed rack location before mounting the chassis in the rack.
- Before using a particular rack, check for obstructions (such as a power strip) that could impair rack-mount installation. If a power strip does impair a rack-mount installation, remove the power strip before installing the chassis, and then replace it after the chassis is installed.
- Allow sufficient clearance around the rack for maintenance. If the rack is mobile, you can push it back near a wall or cabinet for normal operation and pull it out for maintenance (installing or moving cards, connecting cables, or replacing or upgrading components). Otherwise, allow 19 inches (48.3 cm) of clearance to remove field-replaceable units.
- Maintain a minimum clearance of 3 inches on the front and back sides of the chassis for the cooling air inlet and exhaust ports, respectively. Avoid placing the chassis in an overly congested rack or directly next to another equipment rack; the heated exhaust air from other equipment can enter the inlet air vents and cause an overtemperature condition inside the router.

> **Caution**
>
> To prevent chassis overheating, never install a Cisco 8600 Series Secure Routers in an enclosed space that is not properly ventilated or air conditioned.

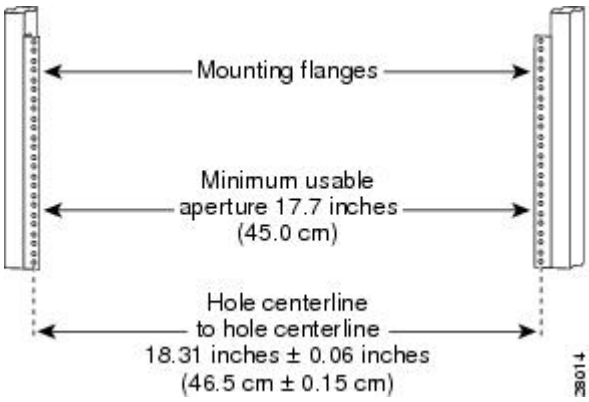
> **Caution**
>
> The recommended torque for rack mount bracket screws is 8 in-lbs.

- Always install heavier equipment in the lower half of a rack to maintain a low center of gravity to prevent the rack from falling over.
- Install and use the cable-management brackets included with the Cisco 8600 Series Secure Routers to keep cables organized and out of the way of the cards and processors. Ensure that cables from other equipment already installed in the rack do not impair access to the cards or require you to disconnect cables unnecessarily to perform equipment maintenance or upgrades.
- Provide an adequate chassis ground (earth) connection for your router chassis.

Verify rack dimensions

Before you install the chassis, measure the space between the vertical mounting flanges (rails) on your equipment rack to verify that the rack conforms to the measurements shown in this figure.

Figure 10: Verify Equipment Rack Dimensions



Procedure

1. Mark and measure the distance between two holes on the left and right mounting rails.

   The distance should measure 18.31 inches ± 0.06 inches (46.5 cm ± 0.15 cm).

> **Note**
>
> Measure for pairs of holes near the bottom, middle, and top of the equipment rack to ensure that the rack posts are parallel.

2. Measure the space between the inner edges of the left front and right front mounting flanges on the equipment rack.

   The space must be at least 17.7 inches (45 cm) to accommodate the chassis that is 17.25 inches (43.8 cm) wide and fits between the mounting posts on the rack.

Install the router

Details procedures for installing the router on a standalone shelf or rack, guiding attachment of front and rear brackets, mounting methods, recommended tools, and proper chassis grounding with step-by-step instructions and supporting tasks.

After unpacking, based on your requirements, you can set up the Cisco 8600 Series Secure Router.

Attach the front rack-mount brackets

Before you begin

Before installing the chassis in the rack, you must install the rack-mount brackets on each side of the chassis.

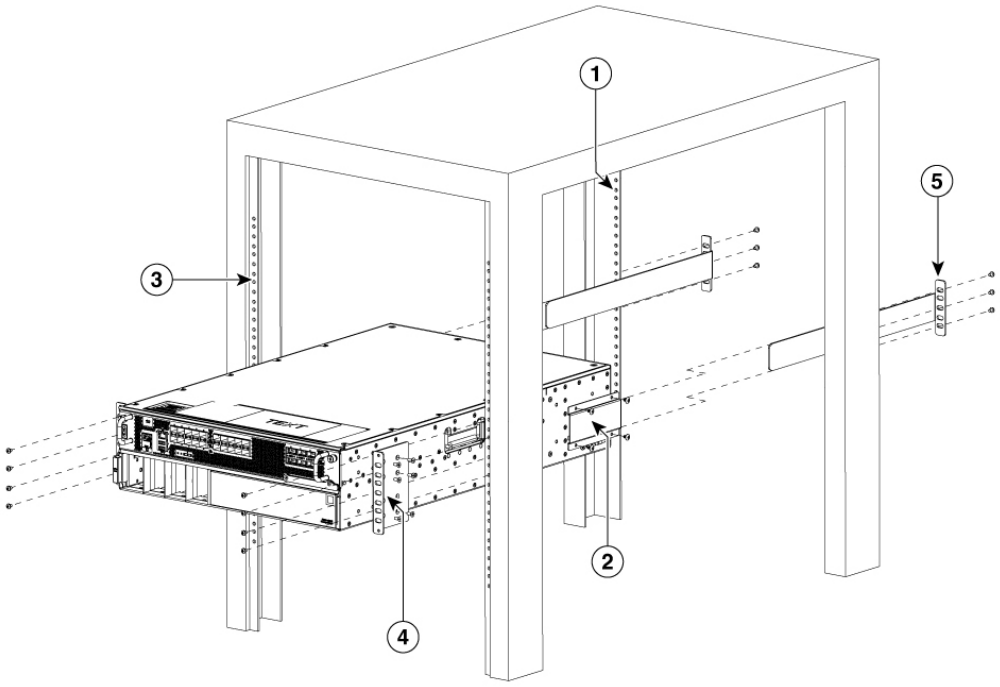
Determine where in the rack you want the chassis to be mounted. If you are mounting more than one chassis in the rack, then start from the bottom up or the center of the rack. The below listed figure shows the brackets attached to the chassis. Depending on the bracket holes you use, the chassis may protrude in the rack.

> **Note**
>
> The cable-management brackets are attached to the chassis after you install the chassis rack-mount brackets on the chassis and mount the chassis in the rack.

Procedure

1. Locate the threaded holes on the side of the chassis. Ensure that you hold the front rack-mount bracket with the ear and holes facing outward and towards the front of the chassis.

   Figure 11: Attaching the Front and Rear Rack-Mount Brackets to the Cisco 8600 Series Secure Routers



1	Rear rack equipment rail	2	Rear rack mount bracket and screws
3	Front rack equipment rail	4	Front rack mount bracket and screws
5	Rear extension brackets		

2. Position the front rack-mount bracket top hole with the chassis, first top hole behind the side vent holes.
3. Insert and tighten the black screws on one side.
4. Repeat Step 1 through Step 3 on the other side of the chassis. Use screws to secure the rack-mount brackets to the chassis.

> **Note**
>
> Use the screw specified in the Bill of Materials (BOM).

Attach the rear rack-mount brackets

Before you begin

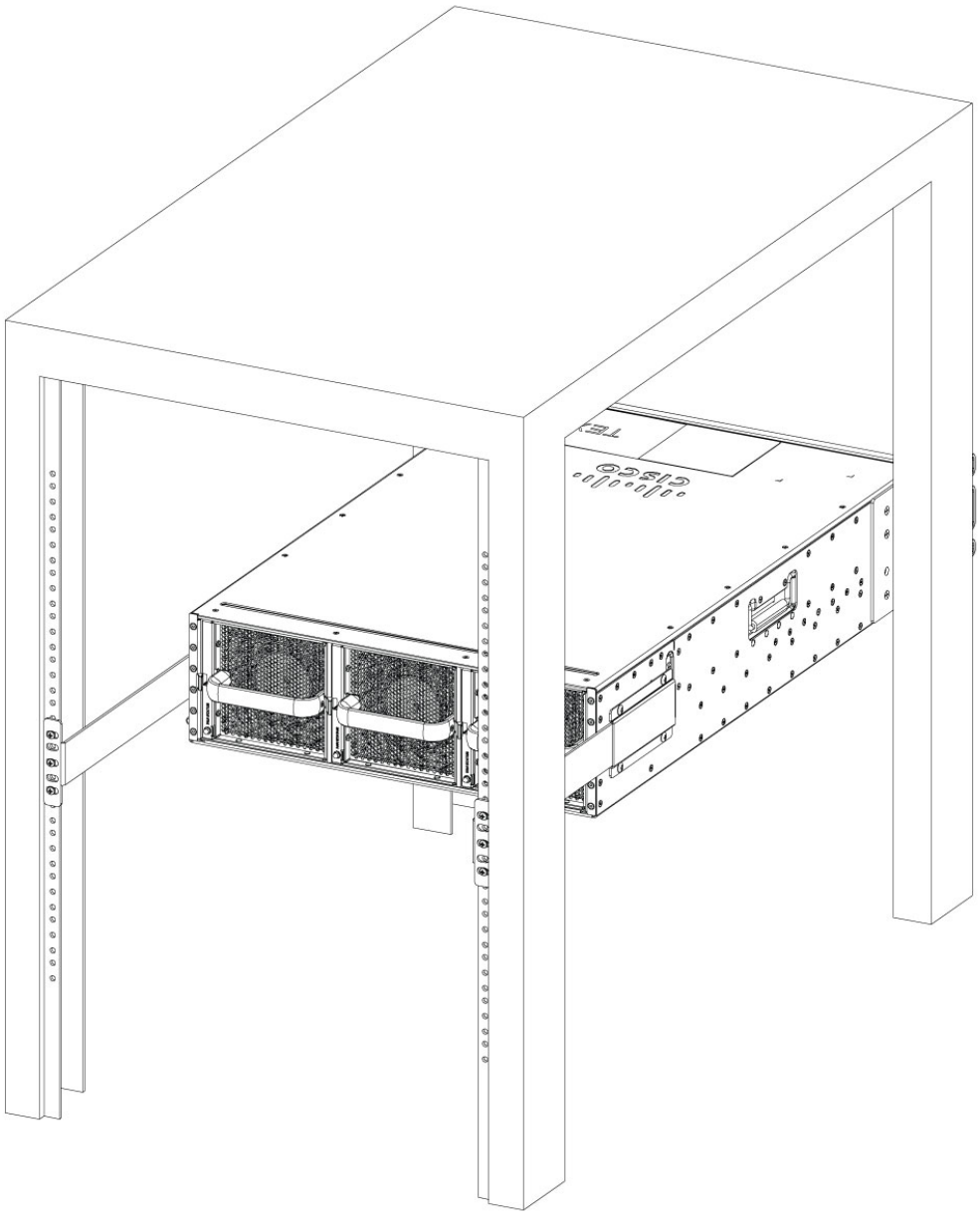
Before installing the chassis in a four-post rack, you must install the rear rack-mount brackets on each side of the chassis.

Determine where in the rack you want the chassis to be mounted. If you are mounting more than one chassis in the rack, then start from the bottom up or the center of the rack. The figure below shows the brackets attached to the chassis.

Procedure

1. Locate the threaded holes on the side of the chassis. Ensure that you hold the rear rack-mount bracket with the ear and holes facing outward and towards the rear of the chassis.

   Figure 12: Rear Rack-Mount Brackets attached to the chassis



2. Position the rear rack-mount bracket with the chassis.
3. Insert the black screws on one side. Do not fully tighten the screws.

   The rear bracket holes are slotted to allow for adjustment. Do not fully tighten the screws until the chassis is installed in the four-post rack.

4. Repeat Step 1 through Step 3 on the other side of the chassis. Use black screws to secure the rack-mount brackets to the chassis.

Mount the router in the rack

After installing the rack-mount brackets on the chassis, mount the chassis by securing the rack-mount brackets to mounting strips in the rack using the screws provided. Because the rack-mount brackets support the weight of the entire chassis, ensure that you use all the screws to fasten the rack-mount brackets to the rack posts.

Four-post rack installation

Procedure

1. (Optional) Install a shelf in the rack to support Cisco 8600 Series Secure Routers . If you use a shelf, it helps support the chassis while you secure it to the rack.

   Note

   If you are using a shelf, place the chassis on the shelf and slightly raise the front of the chassis to align the mounting bracket holes with the rack post holes while allowing the bottom of the chassis to rest on the shelf.

2. With two people, lift the chassis into position between the rack posts.
3. Position the chassis until the rack-mounting flanges are flush against the mounting rails on the rack.

   Note

   Use the second hole up from the bottom of the rack-mount bracket and the second hole down from the top of the rack-mount bracket. This will make it easier to attach the cable-management bracket to the chassis in the equipment rack.

4. Hold the chassis in position against the mounting rails while the second person finger-tightens a screw to the rack rails on each side of the chassis.
5. Finger-tighten screws to the rack rails on each side of the chassis.
6. Tighten all the screws on each side to secure the chassis to the equipment rack.
7. Use a level to verify that the tops of the two brackets are level, or use a measuring tape to verify that both brackets are the same distance from the top of the rack rails.

Attach the cable management bracket

The cable management brackets should be mounted to each rack-mount bracket on the chassis to provide cable management to both sides of the chassis (parallel with card orientation). These brackets are screw-mounted to the rack-mount brackets to allow easy installation and removal of cables.

The cable-management brackets for the Cisco 8600 Series Secure Routers contain one independent cable-management U-type feature with two screws for each bracket.

Note

Make certain that the cable-management bracket "U" feature is facing upwards when you attach it to the Cisco 8600 Series Secure Routers.

Procedure

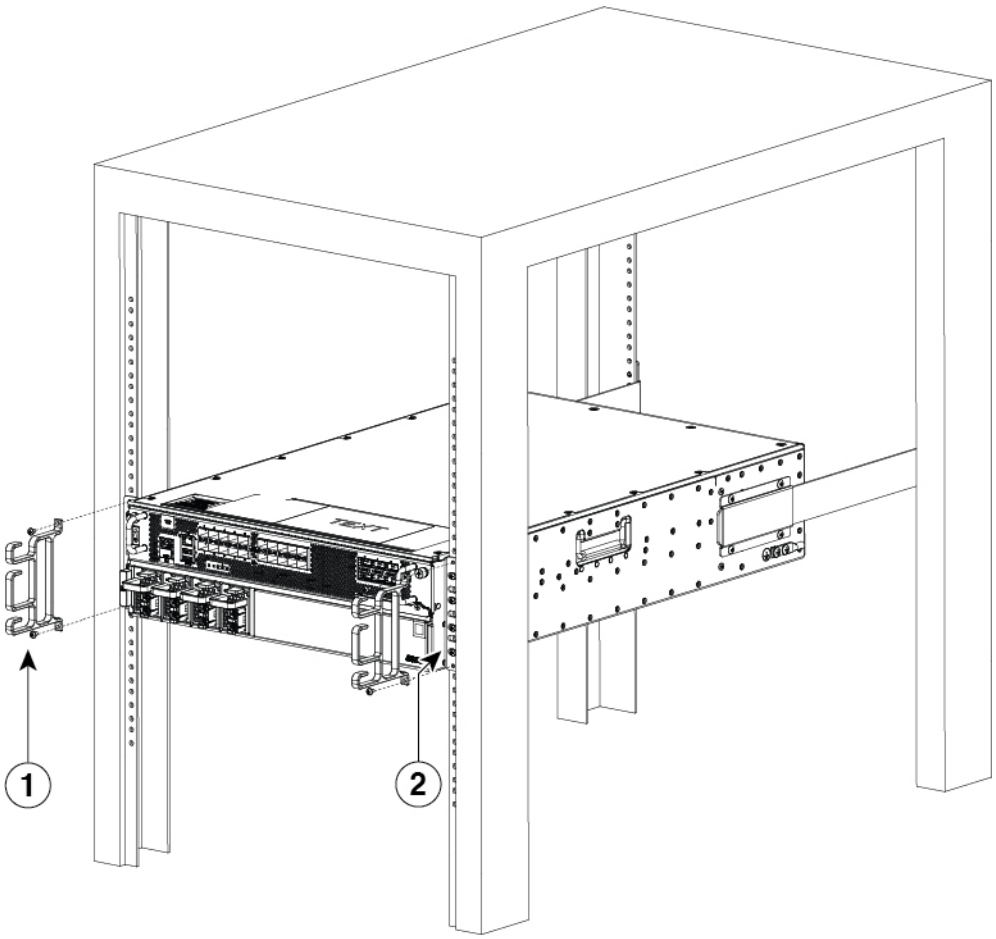
1. Align the cable-management bracket to the rack-mount bracket on one side of the Cisco 8600 Series Secure Routers. The cable-management bracket aligns to the top hole of the chassis rack-mount bracket.
2. Using a Phillips screwdriver, insert one screw through the top screw hole of the cable-management bracket and into the chassis rack-mount bracket and tighten the screw.

   Note

   Use the package of screws that came with your chassis containing four screws.

   This figure shows where to attach the cable-management brackets to the Cisco 8600 Series Secure Routers.

   Figure 13: Attaching the Cable-Management Brackets to C8650-G2



1	Front cable-management bracket.
2	Front-rack mount bracket.

3. Insert a screw through the hole in the cable management bracket and tighten using a Phillips screwdriver.
4. Repeat Step 1 through Step 3 for the other side of the chassis.

Chassis ground connection

Connecting the Cisco 8600 Series Secure Routers chassis to ground is required for all DC powered installations and any AC powered installation where compliance with Telcordia grounding requirements is necessary.

> **Warning**
>
> This equipment must be grounded. To reduce the risk of electric shock, never defeat the ground conductor or operate the equipment in the absence of a suitably installed ground conductor. Contact the appropriate electrical inspection authority or an electrician if you are uncertain that suitable grounding is available.

Before you connect power or turn on power to your chassis, you must provide an adequate chassis ground (earth) connection for the chassis. A chassis ground connector is provided on Cisco 8600 Series Secure Routers. There is a stud on the rear left side of the chassis.

When customer properly install the chassis in a grounded rack, the switch is grounded because it has a metal-to-metal connection to the rack. Alternatively, customer can ground the chassis by using a customer-supplied grounding cable that meets your local and national installation requirements.

For U.S. installations, 6-AWG wire is recommended. Connect the grounding cable to the chassis with a grounding lug and to the facility ground.

Caution

The grounding wire should always be the first to be installed or connected and the last to be removed or disconnected.

Note

An electrical conducting path shall exist between the product chassis and the metal surface of the enclosure or rack in which it is mounted, or to a grounding conductor. Electrical continuity shall be provided by using thread-forming type mounting screws that remove any paint or non-conductive coatings, and establish a metal-to-metal contact. Any paint or non-conductive coatings shall be removed on the surfaces between the mounting hardware and the enclosure or rack. The surfaces shall be cleaned and an antioxidant applied before installation.

Have the recommended tools and supplies available before you begin this procedure.

Recommended tools and supplies

These tools, equipment, and supplies are necessary to connect the system ground to the chassis:

- Phillips screwdriver
- 3.5-mm flat blade screwdriver (Phoenix # 1205053 or equivalent 3.5-mm flat blade)
- Dual-lug chassis ground component
- Grounding wire

Attach a chassis ground connection

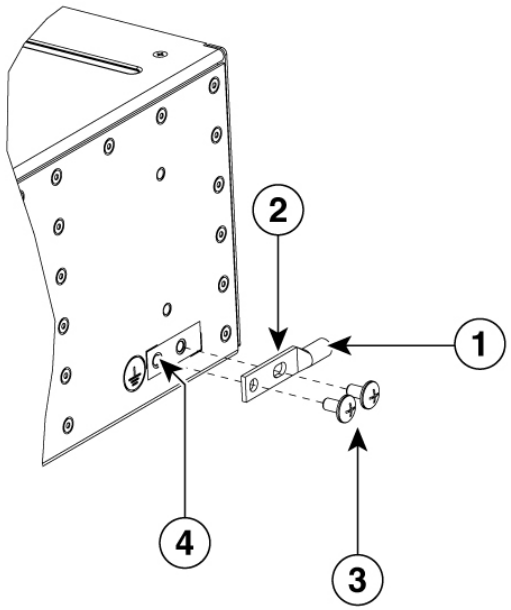
Procedure

1. Use the wire stripper to strip one end of the AWG #6 wire approximately 0.75 inches (19.05 mm).
2. Insert the AWG #6 wire into the open end of the grounding lug.
3. Use the crimping tool to carefully crimp the wire receptacle around the wire. This step is required to ensure a proper mechanical connection.
4. Locate the chassis ground connector on the side of your chassis.
5. Insert the two screws through the holes in the grounding lug.

> **Note**
>
> Use the screw specified in the Bill of Materials (BOM).

This figure shows how to attach a grounding lug to the chassis ground connector.

Figure 14: Attaching a Grounding Lug to the C8650-G2 Chassis Ground Connector



1	Chassis ground connector holes	3	Ground screws
2	Grounding lug	4	Chassis ground connector holes

6. Use the Number 2 Phillips screwdriver to carefully tighten the screws until the grounding lug is held firmly to the chassis. Do not over tighten the screws.
7. Connect the opposite end of the grounding wire to the appropriate ground location on the rack grounding point at your site to ensure an adequate chassis ground.

Connect the power cables

Explains how to connect power and management cables for the router, including console port and USB 3.0 console interface connections, as well as management ethernet cable procedures for reliable device operation.

Follow these guidelines while connecting any external cable to the Cisco 8600 Series Secure Routers:

- To reduce the chance of interference, avoid crossing high-power lines with any interface cables.
- Verify all the cabling limitations (particularly distance) before powering on the system.

Connect the console port cables

The router uses RJ-45 port to attach a console terminal. The router has an asynchronous serial (EIA/TIA-232) RJ-45 console port labeled CON on its front panel. You can connect this port to most types of video terminals with a console cable.

A crossover cable reverses pin connections from one end to the other. In other words, it connects pin 1 (at one end) to pin 8 (at the other end), pin 2 to pin 7, pin 3 to pin 6, and so on. You can identify a crossover cable by comparing the two modular ends of the cable. Hold the cable ends in your hand, side-by-side, with the tabs at the back. Ensure that the wire connected to the outside (left) pin of the left plug (pin 1) is the same color as the wire connected to the outside (right) pin of the right plug (pin 8).

The console port is an asynchronous serial ports; devices connected to this ports must be capable of asynchronous transmission. Baud rates for the RJ-45 console port are 1200, 2400, 4800, 9600, 19200, 38400, 57600, and 115200 bps.

Before connecting to the console interface on the router using a terminal or PC, perform these steps:

Procedure

1. Before connecting a terminal to the console port, configure the terminal to match the chassis console port as follows: 9600 baud, 8 data bits, no parity, 1 stop bits (9600 8N1).
2. Connect one end of the RJ-45 cable to the serial RJ-45 console port (CON) .

   Note

   For information about how to change the default settings to meet the requirements of your terminal or host, see the *Cisco IOS Terminal Services Configuration Guide*.

3. (Optional) After you establish normal router operation, you can disconnect the terminal.

Connect to the USB 3.0 console interface

The USB 3.0 interface connects directly to the USB connector of a PC using a USB Type A to 5-pin micro USB Type-B cable. The USB 3.0 interface supports full speed (12Mbps) operation. The interface does not support hardware flow control.

Note

- Always use shielded USB cables with a properly terminated shield. The USB serial console interface cable must not exceed 3 meters in length.
- Only one interface can be active at a time. When a cable is plugged into the USB 3.0 interface, the RJ-45 port becomes inactive. Conversely, when the USB cable is removed from the USB 3.0 interface, the RJ-45 port becomes active.
- 4-pin micro USB Type-B connectors are easily confused with 5-pin micro USB Type-B connectors. Note that only the 5-pin micro USB Type-B is supported.

The default parameters for the interface are 9600 baud, 8 data bits, no parity, and 1 stop bit.

For operation with a Microsoft Windows OS version older than Windows 7, the Cisco Windows USB Console Driver must be installed on any PC connected to the interface. If the driver is not installed, prompts guide you through a simple installation process.

The Cisco Windows USB Console Driver allows plugging and unplugging the USB cable from the interface without affecting Windows HyperTerminal operations. No special drivers are needed for Mac OS X or Linux.

Management ethernet port cable connection

Procedure

1. Insert an Ethernet RJ-45 cable into the MGMT port.
2. Insert the other end of the RJ-45 cable to your management device or network.

4 Remove and Replace Field Replaceable Units

Topics:

- Install and remove AC power supplies
- Install and remove DC power supplies
- Remove and replace USB 3.0 interface
- Remove and replace the fans
- Installation and removal of air filter
- Install and remove SFP and SFP+ modules
- Remove and replace the battery
- Repack the router

Install and remove AC power supplies

Details the process of installing and removing AC power supplies, including step-by-step instructions for both actions to ensure safe and efficient power supply replacement.

This section describes how to install and remove AC power supplies in Cisco 8600 Series Secure Routers.

Install AC power supplies

> **Note**
>
> Do not install the power supplies with the chassis cover off.

Procedure

1. Ensure that the chassis power switch on the chassis is in the Standby position.

   > **Note**
   >
   > It is not required to place the chassis power switch in the Standby position if you want to hot-swap a single power supply.

2. Insert the power supply module into the appropriate slot(s), making sure that the retention latch is firmly latched. You can verify that the power supply module is firmly latched by gently pulling the power supply handle.
3. Insert the power supply cables firmly into the power supplies.

   > **Note**
   >
   > Ensure that both power supplies are inserted firmly and the power cords are in place.

4. If you have changed the chassis power switch to the Standby position in Step 1, press the power switch to the On position.

   The power supply LEDs are illuminated (green).

Remove AC power supplies

Procedure

1. Ensure that the chassis power switch is in the Standby position.

> **Note**
>
> It is not required to place the chassis power switch in the Standby position if you want to hot-swap a single power supply.

2. Unplug the power cable from the power supply.
3. Press the retaining latch towards the pull handle, grasp the handle with one hand, and pull the power supply out of the slot while supporting the weight of the power supply with the other hand.
4. Repeat these steps if it is required to remove the other AC power supply.

Install and remove DC power supplies

Explains the installation and removal of DC power supplies, covering procedures for installing DC input power supplies, wiring the DC input power source, and safely removing DC power supplies.

This section describes how to install and remove DC power supplies in Cisco 8600 Series Secure Routers.

Install DC input power supplies

> **Warning**
>
> To reduce risk of electric shock or personal injury, disconnect DC power before removing or replacing components or performing upgrades.

> **Note**
>
> Do not install the power supplies with the chassis cover off.

This section describes how to install the DC power supply input power leads to the DC input power supply. Before you begin, read these important notices:

- The color coding of the DC input power supply leads depends on the color coding of the DC power source at your site. Ensure that the lead color coding you choose for the DC input power supply matches the lead color coding used at the DC power source and verify that the power source is connected to the negative (-) terminal and to the positive (+) terminal on the power supply.
- Ensure that the chassis ground is connected on the chassis before you begin installing the DC power supply. Follow the steps provided in the *Chassis Ground Connection* section.
- For DC input power cables, the wire gauge is based on the National Electrical Code (NEC) and local codes for 26 amp service at nominal DC input voltage (-40/-72 VDC). One pair of cable leads, source DC (-) and source DC return (+), are required for each power distribution unit (PDU). These cables are available from any commercial cable vendor.

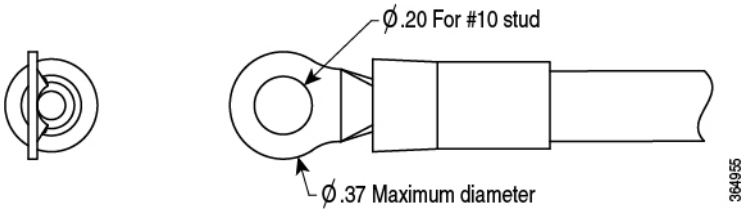
All DC input power cables for the chassis should be 10 gauge wire and cable lengths should match within 10 percent of deviation.

Each DC input power cable is terminated at the PDU by a cable lug, as shown in the below listed figure.

Note

DC input power cables must be connected to the PDU terminal studs in the proper positive (+) and negative (-) polarity. In some cases, the DC cable leads are labeled, which is a relatively safe indication of the polarity. However, you must verify the polarity by measuring the voltage between the DC cable leads. When making the measurement, the positive (+) lead and the negative (-) lead must always match the (+) and (-) labels on the power distribution unit.

Figure 15: DC Input Power Cable Lug



Note

To avoid hazardous conditions, all exposed conductors must be covered. If after placing the protective cover over the DC terminal block, conductors are exposed, these should be insulated to prevent accidental contact.

Wire the DC input power source

Warning

To reduce risk of electric shock, when installing or replacing the unit, the ground connection must always be made first and disconnected last.

If your unit has modules, secure them with the provided screws.

Procedure

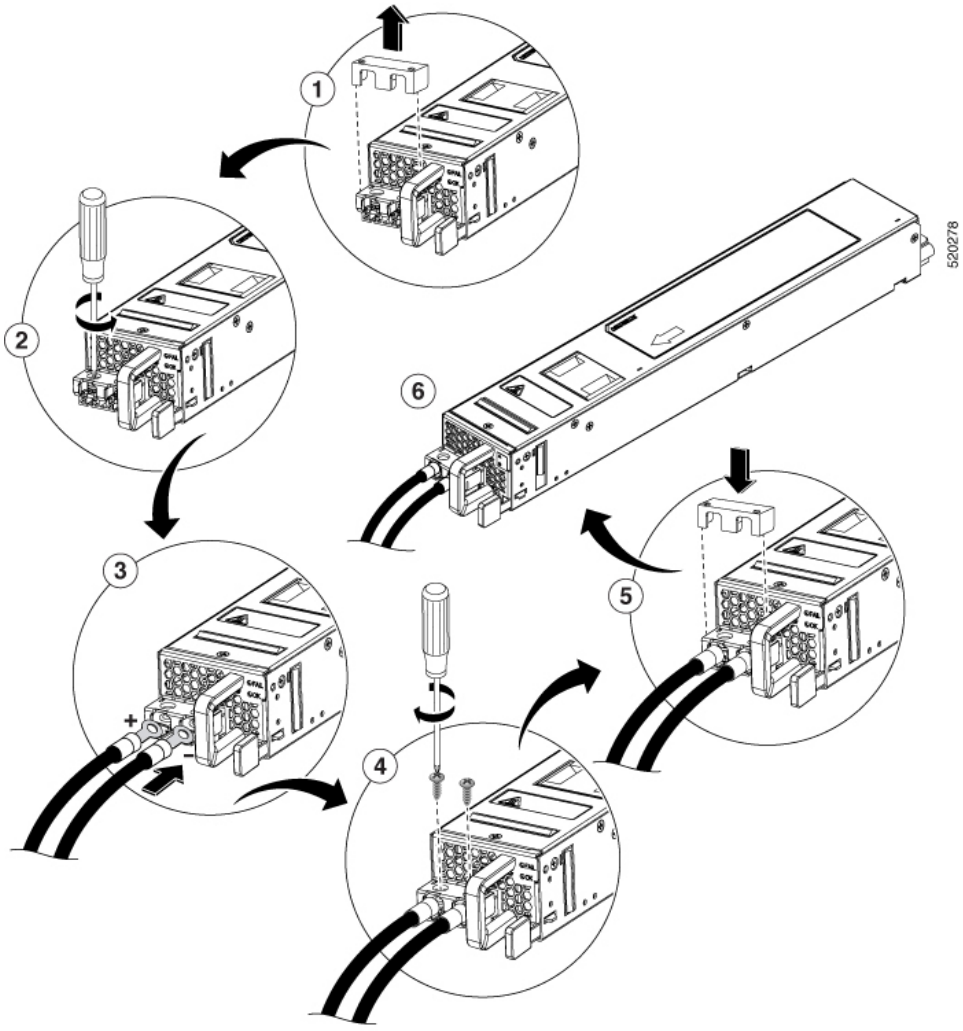
1. Turn off the circuit breaker from the power source.
2. Ensure that the chassis power switch is in the Standby position.

> **Note**
>
> It is not required to place the power switch in the Standby position if you want to hot-swap a single power supply.

3. Use a wire-stripping tool to remove approximately 0.75 inch (19 mm) of the covering from the end of the wire.

   The wire should be stripped so that it extends into the barrel of the lug without extending out of it. Additionally, the insulation should not be inside the barrel.

Figure 16: DC Power Supply Terminal Block Ground Cable Lugs



4. Insert the stripped end of the wire into the open end of the lug.
5. Crimp the wire in the barrel of the lug. Verify that the wire is securely attached to the lug.
6. Place the wire against the terminal block, making sure there is solid metal to metal contact.
7. Secure the lugs to the chassis with two M4 screws. Ensure that the lug and the wire will not interfere with other switch hardware or rack equipment.
8. Replace the snap on cover on the terminal block of the DC power supply.

Remove DC power supplies

The DC power supply has a terminal block that is installed into the power supply terminal block header.

Procedure

1. Turn off the circuit breaker from the power source.
2. Ensure that the chassis power switch is in the Standby position.

> **Note**
>
> It is not required to place the chassis power switch in the Standby position if you want to hot-swap a single power supply.

3. Remove the plastic cover from the terminal block.
4. Unscrew the two terminal block screws on the unit and remove the wires from the power supply.
5. Press the power supply retaining latch towards the pull handle, grasp the handle with one hand, and pull the power supply out of the slot while supporting the weight of the power supply with the other hand.

Remove and replace USB 3.0 interface

Describes the steps required to remove and replace the USB 3.0 interface, providing practical guidance for maintaining or upgrading device connectivity.

The Cisco 8600 Series Secure Router contain one USB 3.0 interface to store configurations or Cisco IOS XE consolidated packages.

To remove and then replace a USB flash memory stick, follow these steps:

Procedure

1. Pull the flash memory stick from the USB port.
2. To replace a Cisco USB Flash memory stick, insert the module into USB port 1. The Flash memory stick can be inserted or removed regardless of whether the router is powered up or not.

Remove and replace the fans

Guides the removal and replacement of fans, including instructions specific to removing fans from the C8650-G2 platform to maintain proper device cooling.

This section describes how to remove and replace the fans in Cisco 8600 Series Secure Routers.

Remove the fans for C8650-G2

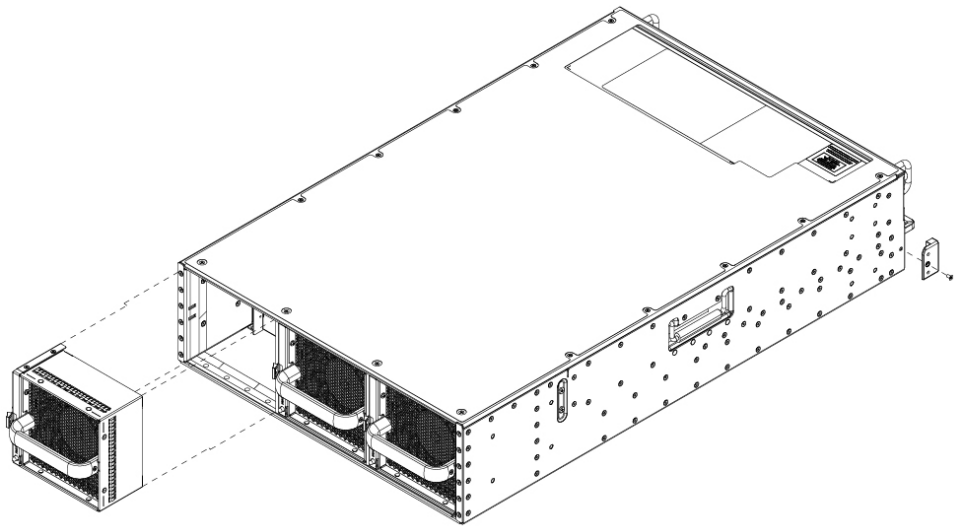
Procedure

1. **Note**

   During fan removal and installation, keep fingers and all other objects away from the chassis opening.

   Position the chassis so that you have the most comfortable access to the chassis to remove the fans. The fans are located at the rear of the chassis.
2. Press the finger tab on the fan latch and pull the fan module straight out to remove the fan.

   Figure 17: Removing the Fans for C8650-G2

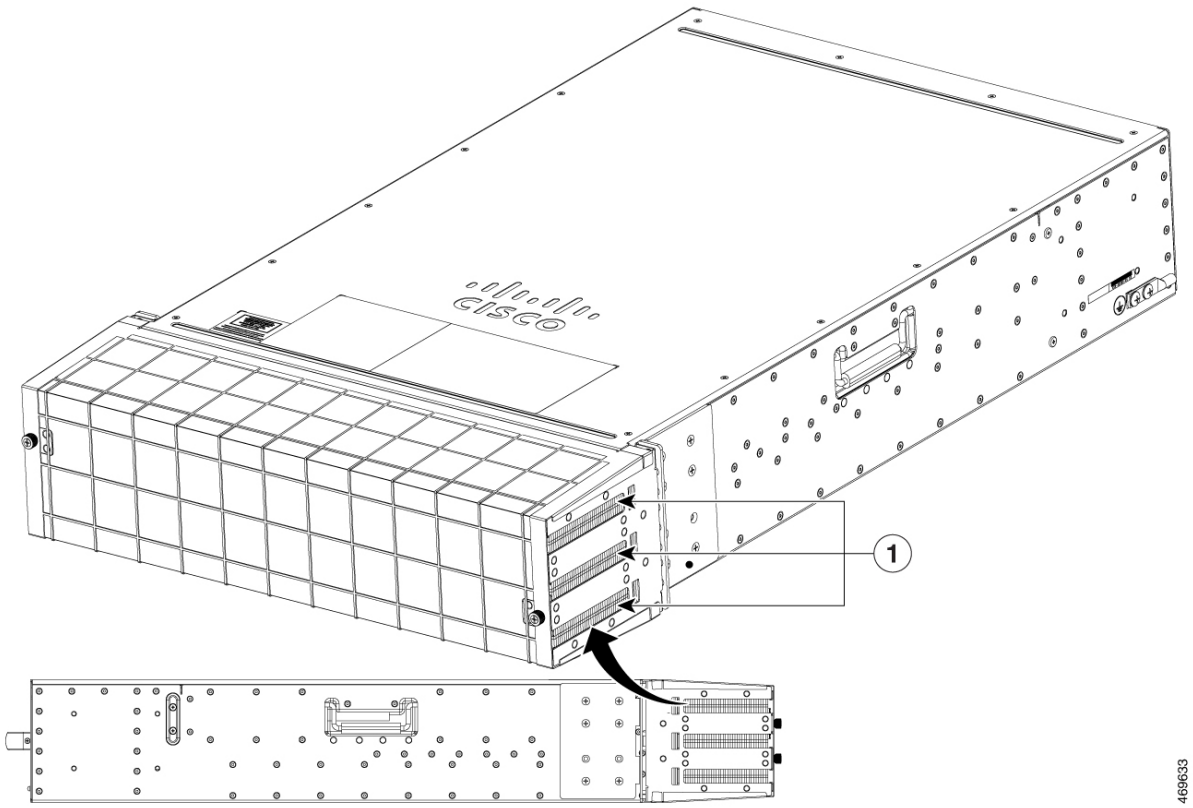


3. Replace the fan module within 3 minutes to maintain proper cooling of the chassis.

Installation and removal of air filter

Explains procedures for installing and removing the air filter, outlining each task to ensure optimal airflow and device longevity.

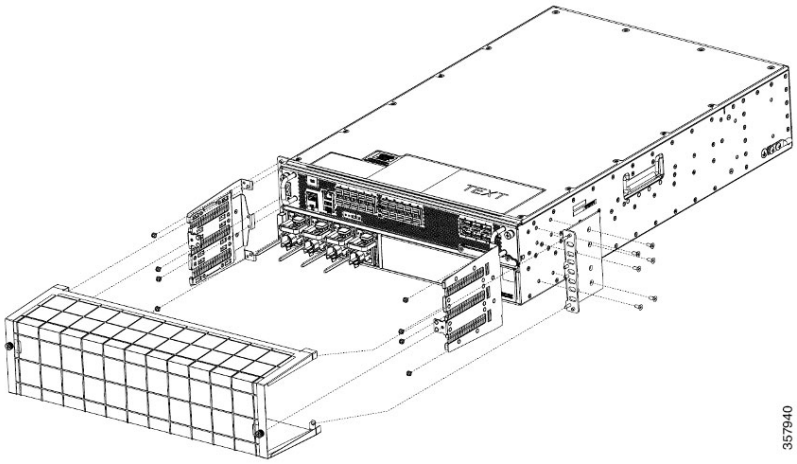
Figure 18: Side View and Front view of Chassis with Air Filter and Cable Management Brackets



1 Cable Exit Areas

Install the air filter

Figure 19: Exploded View of the Air Filter



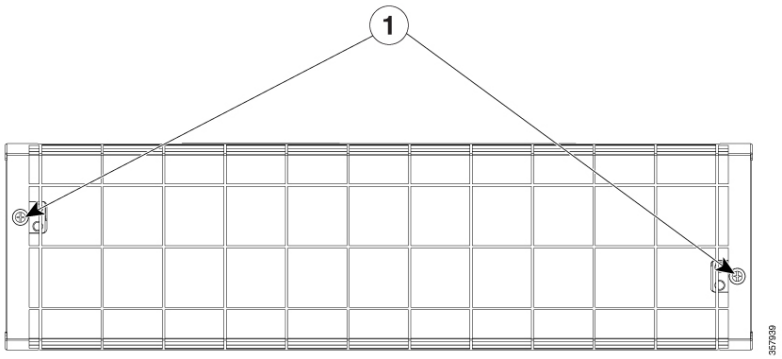
Procedure

1. If the chassis is installed without a filter, it must be powered down and removed from the rack.
2. Install the rack mount brackets as shown in the above figure, using 6 screws per rack mount bracket to mount to the chassis.
3. Align and install the air filter mounting brackets to the front rack mount brackets. Use 4 pan head screws per side.

4. Route the power and data cables to the left and right, through the bristles in the air filter mounting brackets.
5. Power on the chassis.
6. Carefully slide the air filter over the mounting brackets and secure the air filter by tightening the thumb screws on the air filter to the mounting brackets.

Remove the air filter

Figure 20: The Air Filter

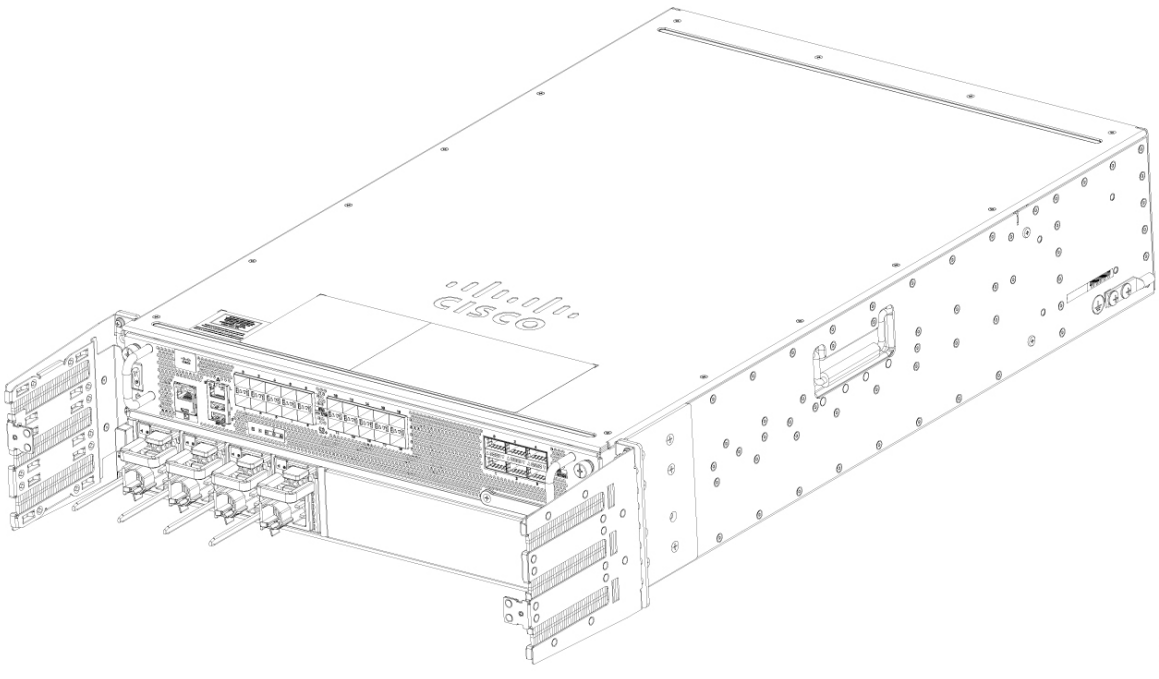


1	Air filter thumbscrews

Procedure

1. Fully loosen the two thumbscrews (as shown in the above figure) on the front of the filter by turning them counter clockwise.
2. Pull the air filter straight off the air filter mounting brackets.
3. Discard the used air filter.

Figure 21: Figure: Chassis With Air Filter Removed



Note

It is recommended that the air filter be change every 3 months. However, examine the air filter once a month (or more often in dusty environments) and replace it if it appears to be excessively dirty or damaged. To comply with Telecordia GR-63-Core standard air filter requirements for NEBS deployments, the air filter must be replaced, not cleaned.

Install and remove SFP and SFP+ modules

Outlines the steps for installing and removing SFP and SFP+ modules, including adherence to laser safety guidelines and proper removal techniques for small form pluggable modules.

Before you begin

See the Cisco Optics-to-Device Compatibility Matrix for a list of supported SFP and SFP+ modules. Use only supported SFP/SFP+ modules on the platform.

Warning

Class 1 laser product. Statement 1008

Note

We recommend that you wait 30 seconds between removal and insertion of an SFP on an interface module. This time is recommended to allow the transceiver software to initialize and synchronise with the standby RSP. Changing an SFP more quickly could result in transceiver initialization issues that disable the SFP

- Do not remove the dust plugs from the SFP and SFP+ modules or the rubber caps from the fiber-optic cable until you are ready to connect the cable. The plugs and caps protect the module ports and cables from contamination and ambient light.
- Removing and installing an SFP and SFP+ module can shorten its useful life. Do not remove and insert any SFP/SFP+ module more often than is necessary.
- To prevent ESD damage, follow your normal board and component handling procedures when connecting cables to the switch and other devices.
- When you insert several SFP and SFP+ modules in multiple ports, wait for 5 seconds between inserting each SFP/SFP+. This will prevent the ports from going into error disabled mode. Similarly, when you remove an SFP and SFP+ from a port, wait for 5 seconds before reinserting it.

Procedure

1. Attach an ESD-preventive wrist strap to your wrist and to an earth ground surface.
2. Find the send (TX) and receive (RX) markings that identify the top of the SFP/SFP+ module.
   On some SFP/SFP+ modules, the send and receive (TX and RX) markings might be shown by arrows that show the direction of the connection.
3. If the SFP/SFP+ module has a bale-clasp latch, move it to the open, unlocked position.
4. Align the module in front of the slot opening, and push until you feel the connector snap into place.
5. If the module has a bale-clasp latch, close it to lock the SFP/SFP+ module in place.
6. Remove the SFP and SFP+ dust plugs and save.
7. Connect the SFP and SFP+ cables.

Laser safety guidelines

Optical Small-Form Pluggable (SFPs) use a small laser to generate the fiber-optic signal. Keep the optical transmit and receive ports covered whenever a cable is not connected to the port.

Warning

Invisible laser radiation may be emitted from disconnected fibers or connectors. Do not stare into beams or view directly with optical instruments. Statement 1051

Warning

Ultimate disposal of this product should be handled according to all national laws and regulations. Statement 1040

Warning

Pluggable optical modules comply with IEC 60825-1 Ed. 3 and 21 CFR 1040.10 and 1040.11 with or without exception for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019. Statement 1255.

To install an SFP module in your device, perform these steps:

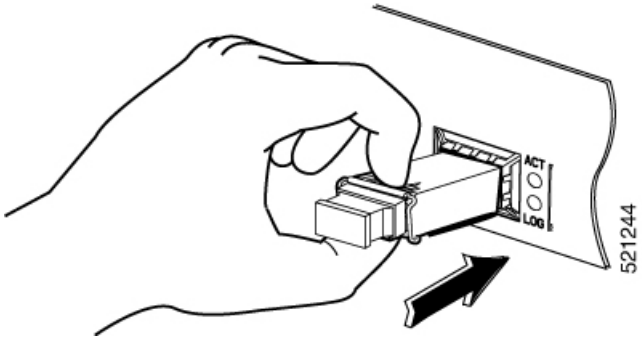
Procedure

1. Read the Safety Warnings section and attach an ESD-preventive wrist strap to your wrist and to the ESD ground connector or a bare metal surface on your chassis.
2. Slide the SFP into the device connector until it locks into position

Tip

If the SFP uses a bale-clasp latch (see Laser Safety Guidelines section, the handle should be on top of the SFP module.

Figure 22: Install a Small-Form Pluggable Module



> **Caution**
>
> Do not remove the optical port plugs from the SFP until you are ready to connect cabling.

3. Connect the network cable to the SFP module.

Remove small form pluggable modules

Follow these steps to remove a Small Form Pluggable (SFP) from the device:

Procedure

1. Read the Safety Warnings section and disconnect the power supply before you perform any module replacement.
2. Disconnect all cables from the SFP.

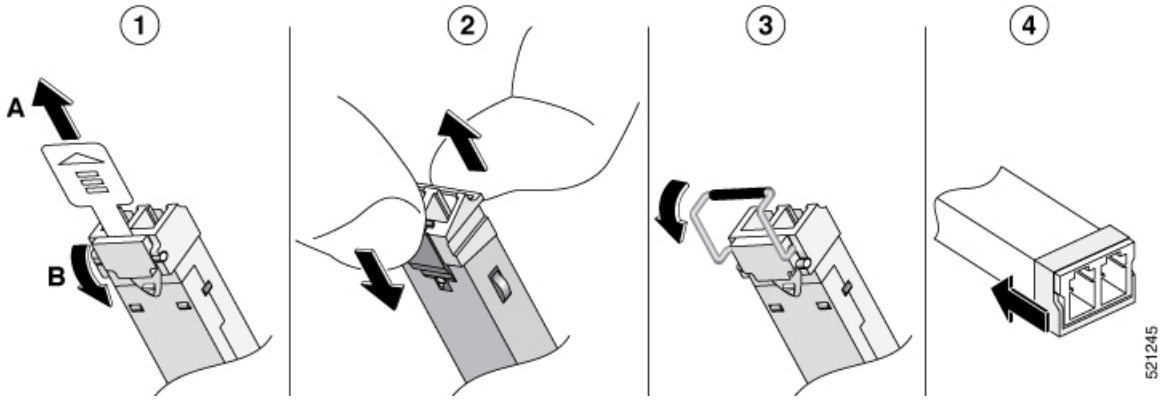
> **Warning**
>
> Invisible laser radiation may be emitted from disconnected fibers or connectors. Do not stare into beams or view directly with optical instruments.

> **Caution**
>
> The latching mechanism used on many SFPs locks the SFP into place when cables are connected. Do not pull on the cabling in an attempt to remove the SFP.

3. Disconnect the SFP latch.

> **Note**
>
> SFP modules use various latch designs to secure the module in the SFP port. Latch designs are not linked to SFP model or technology type. For information on the SFP technology type and model, see the label on the side of the SFP.

Figure 23: Disconnecting SFP Latch Mechanisms



1	Sliding latch	3	Bale-clasp latch
2	Swing and slide latch	4	Plastic collar latch

Tip

Use a pen, screwdriver, or other small straight tool to gently release a bale-clasp handle if you cannot reach it with your fingers.

4. Grasp the SFP on both sides and remove it from the device.

Remove and replace the battery

These instructions are provided in order to comply with the EU Battery Replaceability Requirement.This section describes the procedure to remove and replace the battery from Cisco 8600 Series Secure Router.

These instructions are provided in order to comply with the EU Battery Replaceability Requirement.

- This product was not specifically designed for battery replacement in the field.
- It is recommended that, whenever possible, follow the standard repair or return process for battery issues.

Caution

Always wear an ESD-preventive wrist strap and ensure that it makes good contact with your skin when you remove or install the internal modules. Connect the equipment end of the wrist strap to the metal part of the chassis.

Remove the battery

This task provides instruction on safely removing the battery from the chassis.

Perform this procedure when the battery requires replacement or maintenance on supported hardware.

Note

Follow proper electrostatic discharge (ESD) precautions before handling the battery or device.

Wear an ESD-preventive wrist strap, ensuring that it makes good skin contact. Connect the clip to an unpainted surface of the chassis frame to channel unwanted ESD voltages safely to ground. To guard against ESD damage and shocks, the wrist strap and cord must operate effectively.

Warning

Statement 1015–Battery Handling

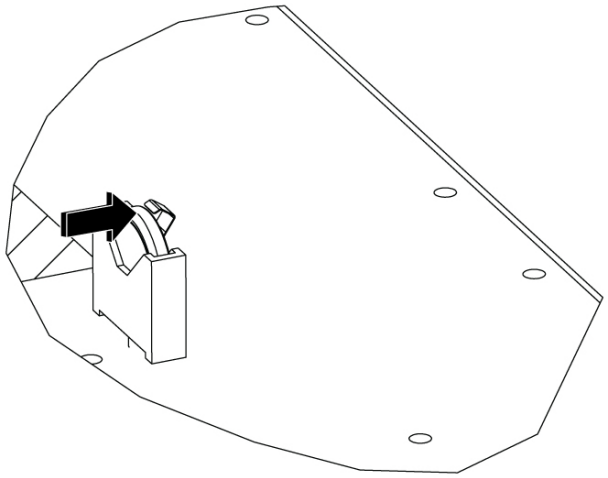
To reduce risk of fire, explosion or leakage of flammable liquid or gas:

- Replace the battery only with the same or equivalent type recommended by the manufacturer.
- Do not dismantle, crush, puncture, use sharp tool to remove, short external contacts, or dispose of in fire.
- Do not use if battery is warped or swollen.

Procedure

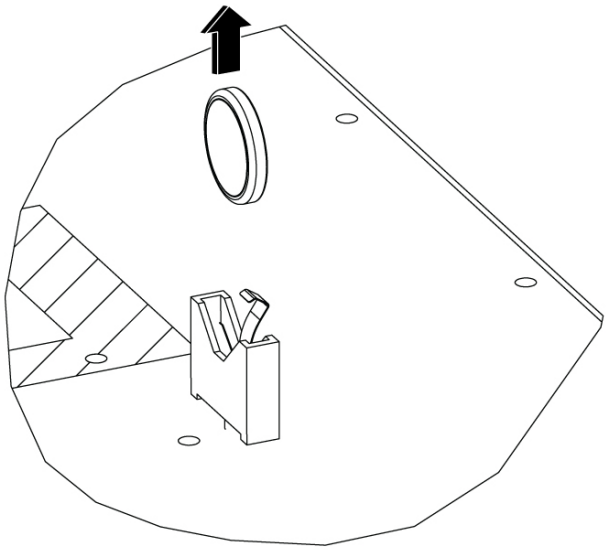
1. Before you perform any module replacement, read the Safety Warnings section and disconnect the power supplies.
2. Pull the retention arm back from the battery.

   Figure 24: Pull back the latch arm



3. Lift the battery out of the holder.

   Figure 25: Remove the battery



Replace the battery

Provides instruction on safely replacing the battery from the chassis.

Replace the battery in the chassis using the safety procedures to ensure proper device functionality.

Note

Follow proper electrostatic discharge (ESD) precautions before handling the battery or device.

Before you begin

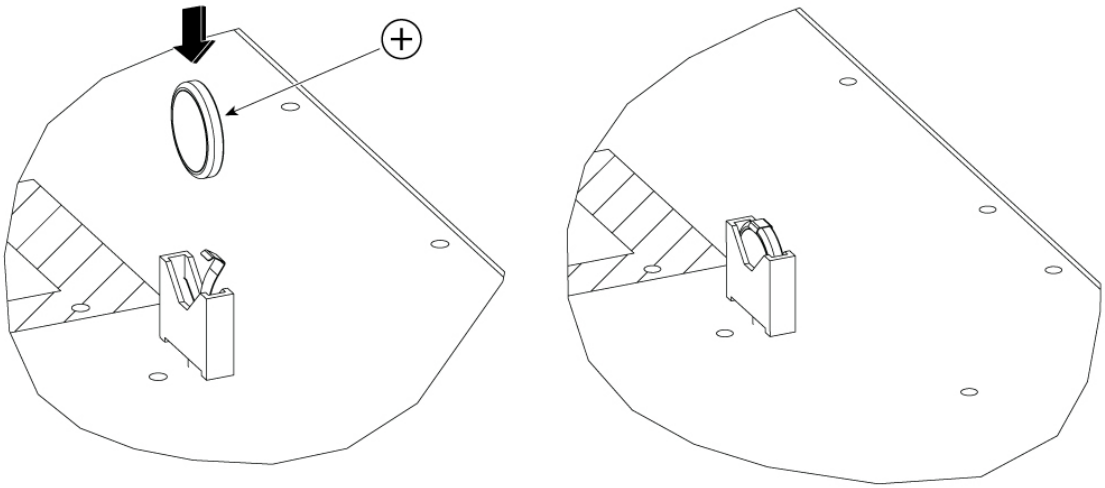
The battery is a CR2032 type. Batteries should only be replaced with the same or an equivalent type recommended by the manufacturer.

- RENATA 100272
- VARTA MICROBATTERY INC 06032 101 501
- Panasonic CR-2032L/BN

Procedure

1. Read the Safety Warnings instructions and disconnect the power supplies before you perform any module replacement.
2. Orient the battery with positive polarity as shown.
3. Push the battery down into the holder. Once installed the latch will snap back over the battery to keep it retained.

Figure 26: Install the battery



Access the battery in C8650-G2

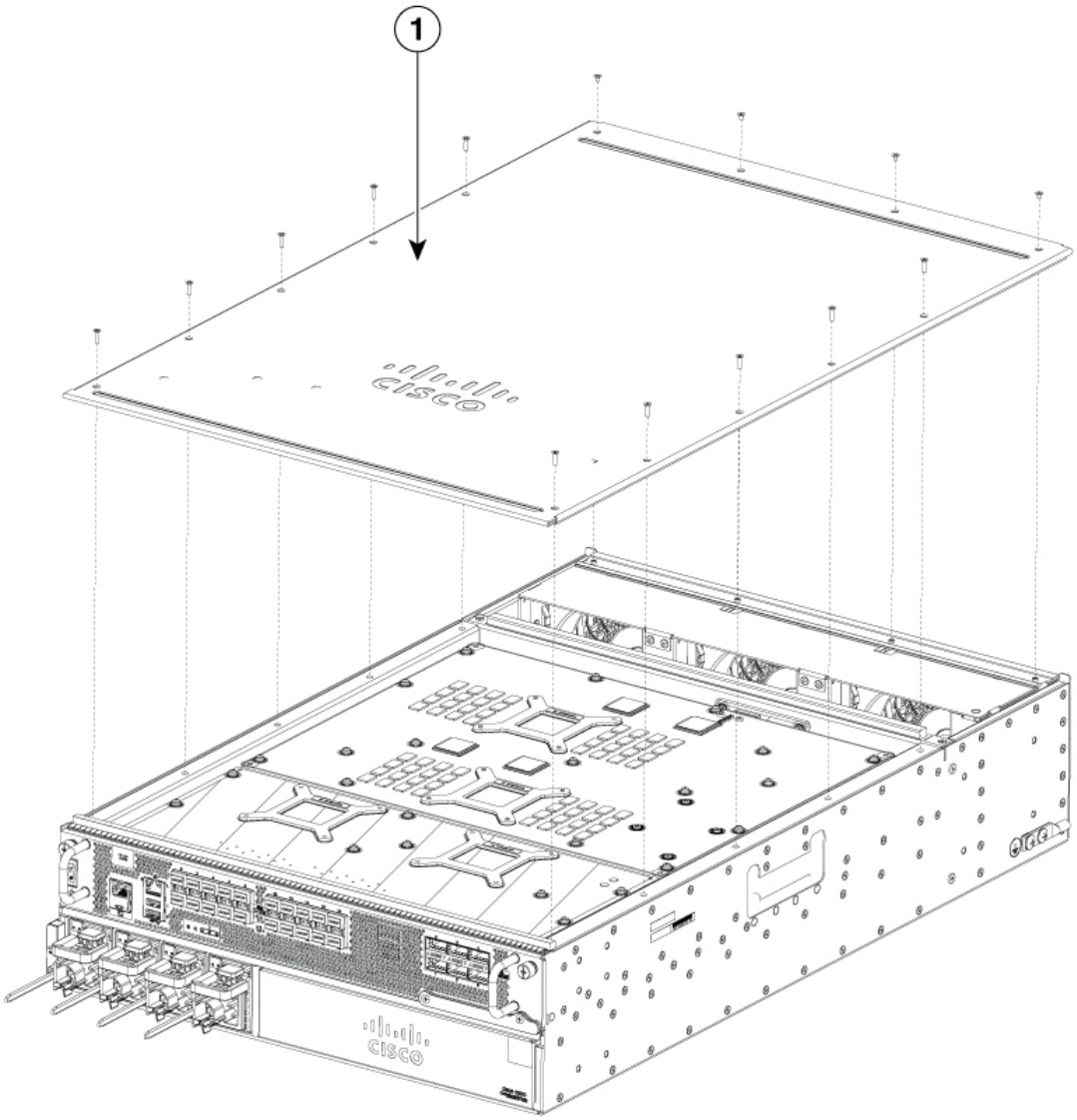
Safely access the battery by powering off the router, removing the chassis cover, removing the Daughterboard, and locating the battery within the hardware assembly.

Use this task when you need to reach the battery inside your router chassis, typically for replacement.

Procedure

1. Ensure the router is powered off and all the power supplies are removed from the chassis.
2. If the router is mounted on a rack, remove the screws from the rack mounting brackets. Remove all screws from the top of the chassis that secure the cover (there are 14 screws).

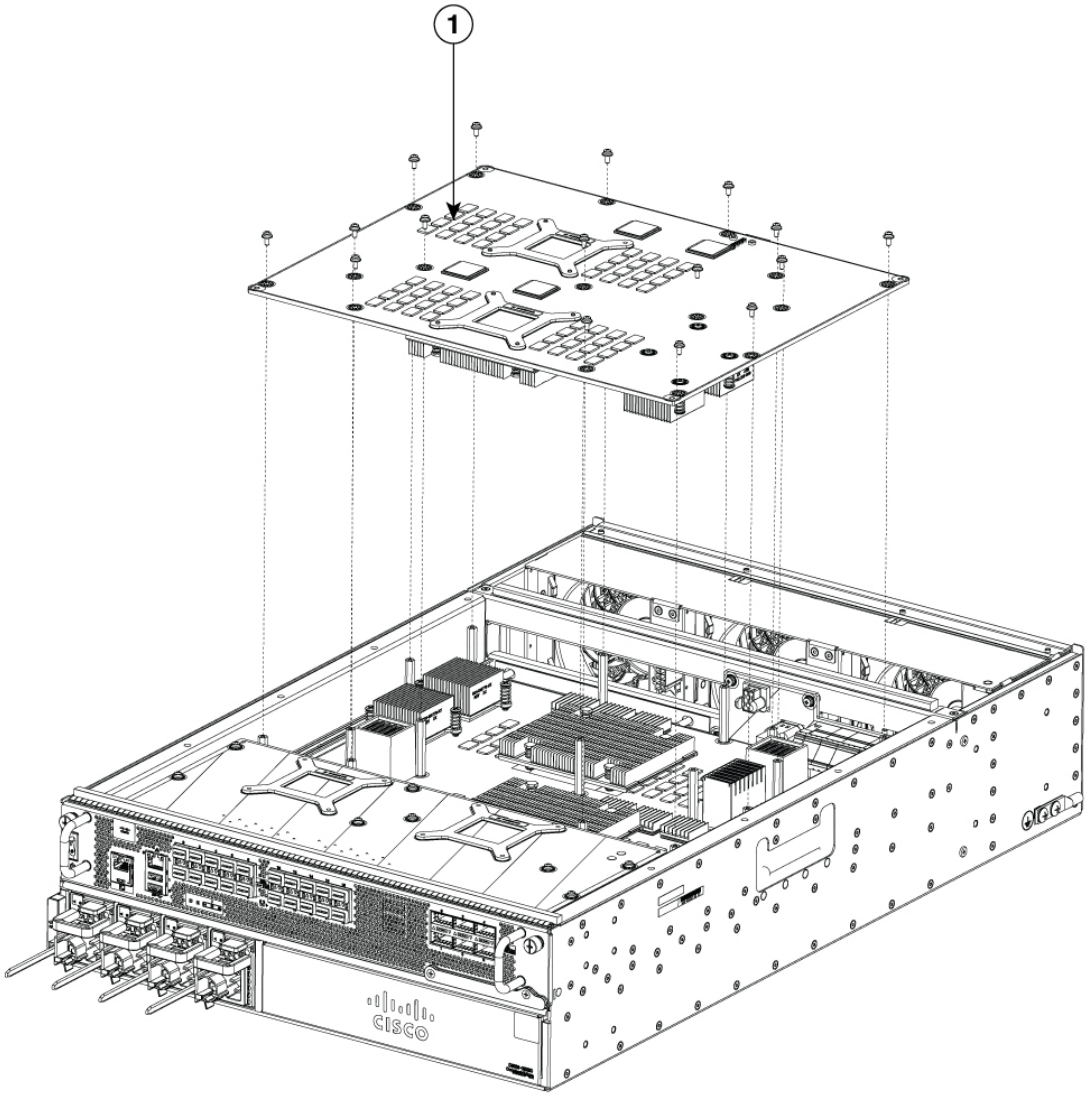
   Figure 27: Remove the chassis cover



1 Top cover off the chassis after removing screws

3. Remove the Daughterboard from the internal standoffs and rest on an antistatic mat (there are 16 screws).

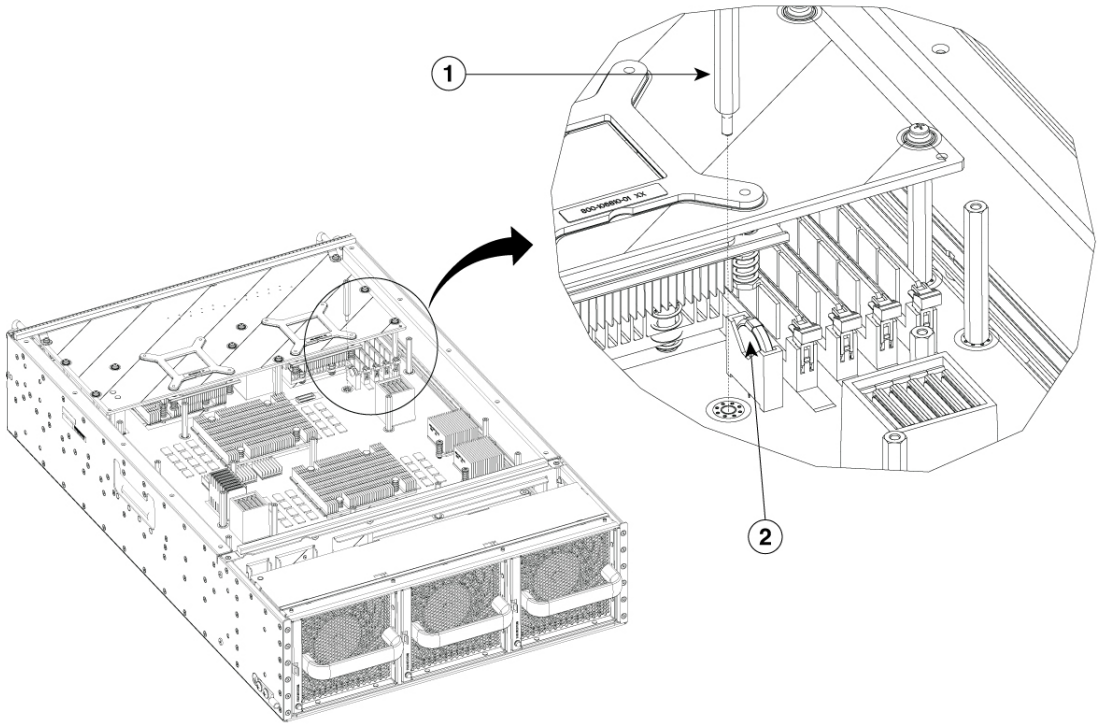
Figure 28: Remove the Daughterboard



1	Daughterboard off standoffs after removing screws

4. Remove the hex standoff near the battery.

Figure 29: Battery location



1	Hex screw removed
2	Battery location

Note

Chassis assembly is the reverse of the disassembly procedure.

Repack the router

Provides guidance for properly repacking the router, ensuring all components are secured for safe storage or transport.

If your system is damaged, you must repack it for return shipment.

Before you return the router or move the router to a different location, repack the system using the original packaging material.